Supplementary material for "Where does the melody go? Measuring and steering melody and lyrics inside an open song model"

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This supplement gives the full account of every study summarised in the article: the model and hooks, the measuring tools and their test on known notes, each experiment with all registered labels, contrasts and post-hoc checks, the work under way, the custody of the registrations, every deviation, the full tables, compute and the pinned versions. Its text is the earlier full write-up of the project, without its introduction and related work. The experiment names map to the article as follows: E0 is the test of the tools (article Section 3.3); E1, E1b and E3 the melody-loss studies (Section 4); E4 the notes-versus-words sweep (Section 5); E5 the motif study (Section 6); E6, B1, B2 and Run C the lyric clock (Section 7); E7 the registered comparison with a commercial model, which has not been run.

S1 The model and the hooks

S1.1 ACE-Step 1.5 as used here

ACE-Step 1.5 [Gong et al., 2026] has two parts. A language model, which we call the planner, writes a plan of the song as discrete tokens at 5 per second. The tokens are finite scalar quantization codes [Mentzer et al., 2024]; our probe reads them as six digits with 8, 8, 8, 5, 5 and 5 levels. A flow-matching transformer, the renderer, turns the plan, the lyric and a caption into latents of a variational autoencoder (VAE) at 25 frames per second with 64 channels, which the VAE decodes to 48 kHz audio. The renderer works on pairs of latent frames, so its queries step every 80 ms. Its cross-attention reads one packed sequence of encoder states that holds the caption, the lyric tokens

and other conditioning; in an English lyric, the first lyric word sits after a 9-token header. The renderer has 384 cross-attention heads (16 in each of 24 layers), and pairs of query heads share a key and value head.

We use the "turbo" checkpoint, which samples in 8 Euler steps. With the time shift of 3.0 that the ACE-Step 1.5 report gives for the 8-step model [Gong et al., 2026], the steps start, as recorded in bf16, at $t = 1.0$, 0.953, 0.898, 0.832, 0.75, 0.645, 0.5 and 0.301 (`results/gpu_run1/e2_smoke_test.json`). Three of the model's tasks appear in this paper. Text-to-music generates from the caption and the lyric, optionally with given plan tokens. Cover generation gives the renderer the dense 25 Hz latents of a source recording as context; with a strength of $k/8$, the context is held for the first k of the 8 steps, after which ACE-Step hands over to text and lyric conditioning alone. Repaint regenerates one time window of a source and keeps the rest. Runs A, B and C used ACE-Step at commit `ca1e85f` with weights pinned to a fixed revision (Appendix SD); the first GPU run (E1, E1b, E3, E4 and E5 v1) downloaded the weights without recording a revision. All GPU runs used bf16 weights, PyTorch's scaled-dot-product attention and no planner language model. In E1 the plan tokens were computed from audio by the model's own tokenizer; elsewhere no plan tokens were given.

S1.2 Hooks on every denoising step (E2)

The turbo sampler integrates

$$x_t = t\,\epsilon + (1 - t)\,x_0, \qquad v = \epsilon - x_0, \tag{1}$$

so at every call of the renderer its estimate of the clean latent is $\hat{x}_0 = x_t - t\,v$. Our wrapper around the renderer's forward pass (`dacapo/ace.py`) reads $\hat{x}_0$ at each step and can replace it with an edited x_0' by returning $v' = (x_t - x_0')/t$. The same wrapper can swap the conditioning before a call (for example the context latents, or the lyric part of the encoder states part way through sampling), record every step, and call the renderer more than once in a step to combine several conditionings. Two further hooks act inside the cross-attention of chosen layers. A tap recomputes the attention probabilities from the module's own inputs and returns nothing, so the forward pass is untouched; asking ACE-Step for its attention weights instead switches the kernels and changed the velocity (relative L_2 difference 0.016 on CPU in bf16). A bias hook adds a fixed matrix to the additive attention mask of chosen heads before the softmax (`dacapo/attn_bias.py`); the lyric clock of Section S6 is built on it. Apart from the combine mode, no hook adds a model call.

A hook is useful only if doing nothing changes nothing. We checked this before every GPU study.

- An identity edit gave a maximum relative difference of 0.0 against the unhooked run, on CPU and on an NVIDIA L4 GPU. Running the same conditioning twice in a combine step (16 denoiser calls for 8 steps) also gave 0.0. Swapping the lyric part way through changed the output (maximum relative difference 0.53 on CPU, 0.28 on GPU), so the swap reaches the model (`results/cpu/e2_smoke_test.json`, `results/gpu_run1/e2_smoke_test.json`). The first version computed $\hat{x}_0$ in bf16 and was not exact; it now works in float32.
- The bias hook at $\beta = 0$ left the predicted latents bit-identical in a CPU test, both for text-to-music and for cover generation (`results/cpu/runb_cpu_test.json`), and again on the GPU before Run B, where the maximum absolute difference was 0.0 for text-to-music and for a cover at $k = 4$. Every hooked run made 40 hook calls, one per layer for 5 layers and 8 steps (`results/gpu_run3/a0_report.json`).
- With deterministic kernels and a fixed VAE decode chunk, a repeated GPU run and a run in a fresh process gave the same latents and audio (maximum difference 0.0; 24 of 24 hard checks passed, `results/gpu_run2/e5v2_a0.json`). The same seed gives different audio on CPU and GPU, so every confirmatory run was made on one GPU type.

One stage of the first GPU run crashed when two guide renders differed by one latent frame; the swap now crops or pads the swapped context by at most two frames and refuses larger differences.

S2 Measuring melody and words

S2.1 Testing the tools on known notes (E0)

Before any model run we measured the measuring tools on audio where every note is known. Melodies were rendered with FluidSynth [FluidSynth developers, 2023] and the FluidR3_GM soundfont [Wen, 2008]: 9 four-bar melodies at 100 BPM (one fixed tune and 8 random ones) in 18 conditions that vary the lead instrument, the level of a band under it and reverb (part A), and 6 melodies in 5 conditions with $k = 0$, 1, 2 or 4 wrong notes (part B). The two pitch trackers tested, SwiftF0 [Nieradzik, 2025] and FCPE [Luo et al., 2026], are accurate on a solo line and much less so under a loud band.

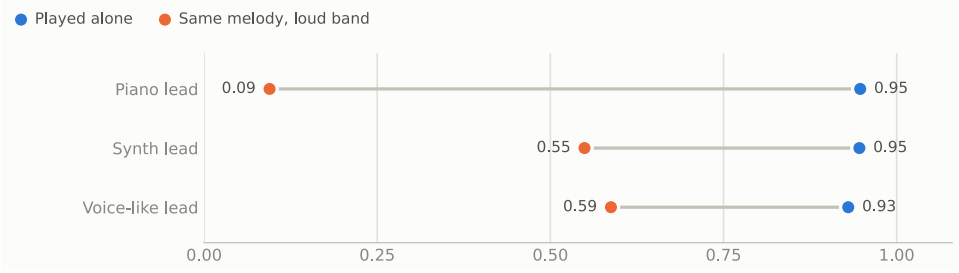


Figure 1: E0: SwiftF0 raw pitch accuracy on a perfectly played melody, alone and under a loud band (MIDI velocity 90). 9 melodies per condition. Source: `results/cpu/e0_ruler.json`.

SwiftF0's raw pitch accuracy on a solo piano lead is 0.947 and falls to 0.095 when the band plays at MIDI velocity 90; for a synthesizer lead it falls from 0.946 to 0.549, and for a voice-like lead ("choir aahs") from 0.930 to 0.587 (Figure 1; `results/cpu/e0_ruler.json`). On a solo line one wrong note is easy to detect. With a voice-like lead over a band, the best of four scores detects one or two wrong notes with an AUC of only 0.65 to 0.78 (6 melodies per condition).

Two rules follow, and the rest of the paper keeps them. A score computed on a full mix is always read next to the same score on the input mix or against a chance level from other melodies. Singing is scored only in clips whose caption asks for no instruments, and in Run B the notes are scored on the vocal stem separated by htdemucs [Rouard et al., 2023].

S2.2 The measures

Pitch-class accuracy (E1). For every frame where the written melody sounds, the dominant chroma bin of the output (above 250 Hz) is compared with the written pitch class, octave ignored, at the best single time lag within ±0.3 s. Each output is also scored against the other melodies at the same tempo; that mismatched score is the chance level, and the matched score minus the mismatched score is the melody information that survived. Note F1 comes from Basic Pitch notes reduced to the top line, scored with mir_eval (50 ms onset tolerance, 50 cents).

Note pitch accuracy (NPA; E4 and B2). A written note counts as sung right if the median SwiftF0 frequency over the central 60% of the note lies within 50 cents of the written pitch, in the written octave. Notes with

too few voiced frames count as wrong. One global lag within ±0.5 s aligns output and score. NPA_v is NPA on the separated vocal stem. Its chance level, NPA scored against the other melodies of the set, was about 0.10 in every B2 arm.

Accuracy against a hypothesis (E5). In the motif study, every possible answer is rendered exactly from symbols. A gap's accuracy against a hypothesis is the share of the hypothesis's notes matched by the median SwiftF0 frequency over the central 60% of the note, within 50 cents, after a global lag search of ±120 ms, leaving out 0.4 s at each edge of the gap. Before any model run, this ruler had to sort renders of every hypothesis correctly at least 95% of the time and detect one wrong note with an AUC of at least 0.9 (kill criterion K0).

Vocal gate. A clip has vocals if its htdemucs vocal stem holds at least 20% of the stem energy, at least 1.0 s of its 50 ms frames reach −40 dBFS, and its peak before normalisation is at least 0.01. The gate asks whether vocals are present, not whether they dominate.

Word recall (WR). Whisper large-v3 [Radford et al., 2023], at a pinned revision, in fp16 with greedy decoding and English set, transcribes the full mix with word time stamps. It is never given the lyric. Lyric words are matched to Whisper's words in order by exact normalised spelling (`dacapo/words.py`), and WR is the share of lyric words matched. The E4 figure counts words regardless of order; on E4 both counts give the same headline (Section S4). As a second opinion in B2 and Run C, a wav2vec 2.0 model fine-tuned with CTC [Baevski et al., 2020], at a pinned revision, decodes the mix greedily, and the same matcher gives CTC WR; Run C also decodes the separated vocal. WR has a chance level too, since the lines are short and full of common words. Scoring each B2 transcript against the other 23 lyrics of the test set gives a mean WR of 0.14 for the clocked recipe, 0.09 for the same guide without the clock and 0.11 with rotated windows, and no clip reaches 0.5 against any other lyric (post-hoc and exploratory, with `dacapo/words.py`; `results/checks/runb_secondary.json`).

Joint score. In B2, a clip's score is $q = \text{gate} \times \min(\text{NPA}_v, \text{WR})$. A clip is *joint* if $q \geq 0.5$, that is, if it has vocals and at least half of the notes and at least half of the words are right, and *strict* if $q \geq 0.75$.

S2.3 What the tools can and cannot see

The pitch trackers do not hear words, and the speech recognisers do not hear pitch. Neither says whether a voice sounds natural or whether the words are clear to a listener. Several failure modes appeared in the data.

- Whisper writes text for wordless audio. In E4, 106 of 504 transcripts had a word error rate above 1, so we report word recall rather than WER. Some transcripts are fluent credit lines unrelated to the audio [Koenecke et al., 2024, Barański et al., 2025]. On vowel-only singing Whisper can also loop ("oh, oh, . . . ") up to its token limit, which slowed scoring by an order of magnitude and distorted a timing estimate in Run B (Section S6).

- Whisper's word timing is itself uncertain. On the 50 evaluable read-out clips of E6, Whisper and a forced aligner (torchaudio's MMS_FA model; Pratap et al., 2024) differ by a median of 0.39 s (`results/gpu_run2/e6_readout.json`).

- The CTC recogniser reads every B2 arm poorly: its mean word recall is 0.05 to 0.18, against 0.07 to 0.50 for Whisper (`results/gpu_run3/b2_result.json`). In Run C it is 0.08 to 0.13 on the mix and 0.09 to 0.14 on the separated vocal, against 0.19 to 0.44 for Whisper (`results/gpu_run4/c1_result.json`). It was trained on read English speech and was not checked on singing here; wav2vec 2.0 models need transfer learning to transcribe sung lyrics well [Ou et al., 2022].

- NPA and WR are scored separately on the same clip. The joint score therefore does not test that the recognised words are sung on the notes that are right. The listening test under way also asks about the melody and the words as two separate questions.

- SwiftF0 makes octave errors on high flute notes, and Basic Pitch often reports a spurious low note near 49 Hz, so its polyphony count cannot measure accompaniment.

- The in-order word matcher is the registered one, but it does not always find the longest in-order match. In B2, a longest-match matcher finds one or two more words on 12 of the 480 Stage T clips; J and the verdict do not change (post-hoc; `results/checks/runb_secondary.json`).

Every result below therefore names its score, and none of them is a judgement of how the output sounds.

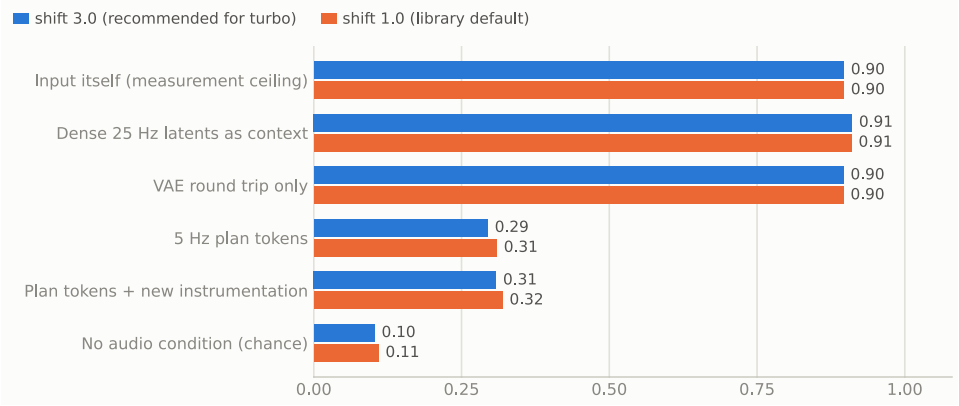


Figure 2: E1: pitch-class accuracy after each path, pooled over 16 melodies and three tempos, at shift 3.0 and 1.0. The no-audio bar rests on only 6 distinct generations per setting. Source: `results/gpu_run1/e1_melody_paths_shift*.json`.

S3 Where the melody is lost

S3.1 Each path through the model (E1)

We rendered 16 random eight-bar melodies, half with a piano lead and half with a flute lead, over block chords, bass and drums, at 80, 120 and 160 BPM. Each clip went through five paths of the model with one seed: a VAE round trip (encode and decode only); the dense 25 Hz latents as context for all 8 steps (ACE-Step's cover task at full strength); the plan tokens that the model's tokenizer computes from the clip's own audio, rendered with the original caption; the same tokens with a caption that asks for different instruments; and text-to-music with no audio condition. The token path is an easy case for the planner, since the tokens come from the target audio itself. It measures what the channel carries for tokens that the model's own tokenizer makes from the audio, from one encoding and one seed; the encoder samples (see below), and tokens written by the planner language model or optimised tokens were not tested. All 480 outputs (16 melodies $\times$ 3 tempos $\times$ 5 paths $\times$ 2 settings of the time shift) were made on an NVIDIA L4; 96 of them are VAE round trips that do not involve the renderer.

With the source latents held for all 8 steps, the output keeps the input's pitch classes; through the plan tokens it does not (Figure 2; Figure 1 of the article). The dense path is close to a reconstruction of the input, so it shows that the renderer can pass a melody through, not that it keeps one while it

also generates new content; Section S4 shows what holding a guide for several steps costs the words. At shift 3.0, pooled over the three tempos, with 95% bootstrap intervals over the 16 melodies, pitch-class accuracy is 0.90 [0.84, 0.94] for the input itself, 0.91 [0.86, 0.95] with dense context, 0.90 after a VAE round trip, 0.29 [0.25, 0.34] through the plan tokens, 0.31 through the tokens with new instrumentation, and 0.10 with no audio condition. Shift 1.0 moves no path by more than 0.02. After subtracting the mismatched chance level, the token path keeps 28% [22%, 34%] of the melody information that the input carries, and dense context keeps all of it (102%). Note F1 tells the same story: 0.78 for the input, 0.72 with dense context, 0.70 after the VAE round trip and 0.12 through the tokens. We found no large tempo effect: the token path's accuracy at 160 against 80 BPM differs by −0.002 [−0.05, 0.04]. The no-audio condition depends only on the caption, the length and the seed, so its 48 outputs per setting contain only 6 distinct generations and its interval means little. The pooled values are means of the per-clip rows in the result files; the intervals were recomputed from those rows with a bootstrap over melodies (10,000 resamples) and agree with the run log to within 0.002.

An earlier CPU run with 4 melodies per tempo gave the tokens 0.51 at 90 BPM. The GPU run did not reproduce it. The two runs differ in VAE precision, random streams and tempo grid, and we do not know which of these explains the gap, so we report the larger GPU run.

S3.2 What the plan tokens hold (E1b)

To locate the loss, we trained linear probes [Alain and Bengio, 2016] to read the pitch class of the lead from each 200 ms frame. The stimuli were 48 four-bar melodies at 120 BPM, each with a piano, flute and violin lead over the same band. Probes learned on 42 melodies and were tested on the other 6 (720 frames, one split). From the plan tokens the probe reaches 0.575, and 0.667 with one-hot features of the six code digits; from the audio latents pooled to the same 5 Hz rate it reaches 0.919; the majority class gives 0.125 (Figure 3). The low frame rate is not the whole reason. On frames where the note does not change, the latent probe reaches 0.982 and the token probe 0.645 (one-hot digits 0.748). On frames with a note change the values are 0.773 and 0.412, and for a lead instrument held out of training 0.846 and 0.504 (`results/gpu_run1/e1b_probe.json`).

The probe and E1 cannot simply be subtracted: they use different stimuli (48 melodies at 120 BPM against 16 over three tempos), different units (probe frames against chroma accuracy) and one test split of 6 melodies, and a

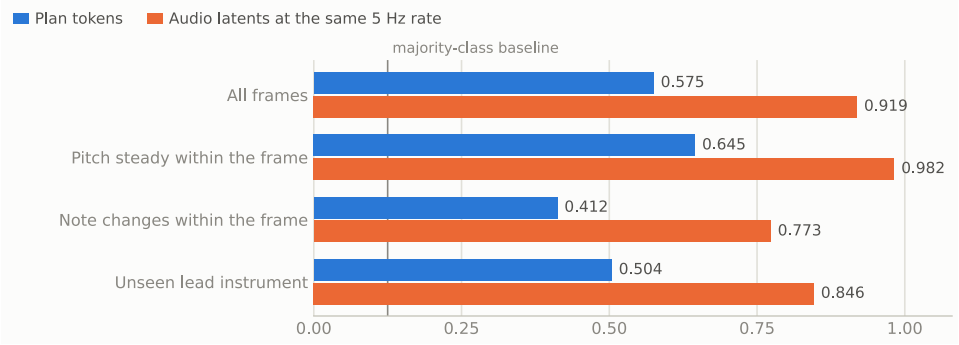


Figure 3: E1b: linear-probe pitch-class accuracy on 6 held-out melodies (720 frames), plan tokens against audio latents pooled to the same 5 Hz rate. Source: `results/gpu_run1/e1b_probe.json`.

linear probe only gives a lower bound on what the tokens hold. Still, a linear probe reads pitch class from the tokens at 0.58 to 0.67, well above the 0.29 that the token path renders. Part of the loss is in the tokenizer, and part happens when the renderer uses the tokens.

The tokens also mix the instrument into the plan. Encoding the same audio twice keeps 84.5% of tokens identical (6 clips), because the encoder samples. Changing only the lead instrument keeps 27.1% (flute) and 27.6% (violin) identical over 48 melodies, about as few as changing the melody (29.1%). These rates count whole clips, including frames after the melody ends, so they understate the change inside the melody. The token file of this run was not saved, so the probe results have no confidence intervals. A CPU run with a different split and a less balanced test set gave the token probe 0.71 (majority baseline 0.29).

S3.3 Pitch in the renderer's latents (E3, exploratory)

If the renderer's latents carry pitch, a melody could be steered there. Chromatic scales in 30 General MIDI instruments were encoded, and a classifier was trained on 24 instruments and tested on the 6 others (2688 test frames). Pitch among 32 classes is read at 0.882 from a 16-dimensional discriminant subspace, 0.880 from 31 dimensions and 0.879 by the nearest class mean in the full 64-dimensional latent, against a chance level of 0.031; pitch class at 0.920 (`results/gpu_run1/e3_readout.json`). A special subspace adds nothing over the full latent. The weakest held-out instrument is choir aahs (0.583), probably because of its slow attack. Replacing the 31 discriminant

coordinates of a melody's latents with those of a target melody, in the VAE alone, moved 0.61 of the frames to the target pitch, against 0.08 without an edit (8 melodies; `results/gpu_run1/e3_edits.json`). A random-subspace control was not run, so this edit is a demonstration, not a finding.

These three results decided the direction of the rest of the project: since the melody survives in the renderer's dense input and latents but not through the plan, control was built inside the renderer.

S4 Notes versus words (E4, exploratory)

E4 asked whether the renderer can sing a given melody with given words when it hears the melody without words. Its settings were not registered before the run, so it is exploratory. The items are 12 lyric lines, one note per one-syllable word (11 to 13 words), with random melodies at 100 BPM in 14 s clips, and an a cappella caption ("a cappella, solo female vocal, no instruments, no drums, dry studio recording, clear lyrics"). Three items were used to try settings and nine were held out. The guide is the melody rendered by FluidSynth as Voice Oohs or as a clarinet. There were 21 settings: no guide; the guide as dense context for the first k of 8 steps, $k \in \{1, 2, 3, 4, 6, 8\}$; and generation started from the guide's latent noised to $t \approx 0.90$, 0.83, 0.75 or 0.64, the SDEdit method [Meng et al., 2022], used here as a baseline. Each setting ran with 2 seeds, 504 generations in all. Notes are NPA on the whole output, which the caption asks to be unaccompanied (E4 used no source separation and no vocal gate), and words are the share of lyric words in Whisper's transcript. At large k a high NPA may therefore come from the wordless guide carried through rather than from singing.

The result is a trade-off with almost nothing in the corner where both are right (Figure 4). Without a guide the model does not know the melody: 0.03 of the notes are at the written pitch, the same as the chance level from mismatched melodies (0.04), while 0.66 of the words are recognised. Every setting that put 0.85 or more of the notes at pitch (nine settings, up to 1.0) kept at most 0.03 of the words. Between the extremes the two move against each other: the Voice Oohs guide held for 3 steps gave 0.40 of the notes and 0.23 of the words, and for 4 steps 0.86 and 0.03. Only 2 of the 378 held-out clips had at least half of the notes and half of the words right (4 of all 504 clips), whether words are counted in order or not. The weakest guide, Voice Oohs released after one step, raised note accuracy by 0.046 over no guide, about half a note per line. Trained systems report the same tension: YingMusic-Singer describes a trade-off between lyric accuracy and melody

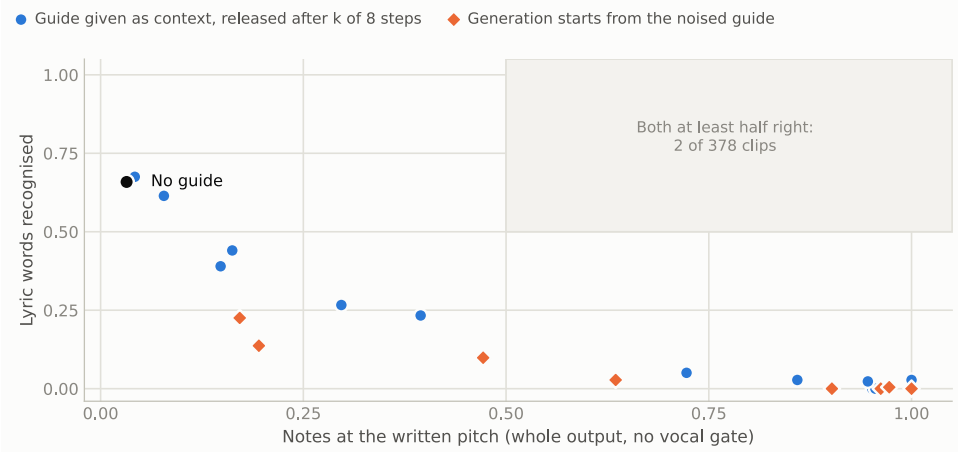


Figure 4: E4: each mark is one guide setting, averaged over 9 held-out melodies × 2 seeds; notes at the written pitch in the whole output (no vocal gate) against lyric words recognised by Whisper. Drawn from the result file with the same code as the repository's figure, without its title. Source: `results/gpu_run1/e4_sweep.json`.

adherence that it reduces with reinforcement learning [Hao et al., 2026], and MPEcho added phoneme conditioning because melody conditioning alone left many lyric errors [Lee et al., 2026].

A working hypothesis came out of these outputs. Without a guide the model sings these lines early and fast (B1 later measured the words starting a median of 3.6 s before their notes). Under a moderate guide, the guide's note times and the model's own lyric timing disagree, and the model resolves the conflict by dropping the words. If the lyric attention could be moved onto the guide's note times, the two signals would agree. That is the idea behind the lyric clock (Section S6). A simpler rival explanation fits E4 equally well: the longer a wordless input is held, the closer the output stays to it, and that input has no words. E4 did not measure how closely the output copies the guide; B2 did, and its guide-copy index was 0.60 for the guide held 4 steps against 0.42 for 3 steps (Appendix SB).

Table 1: E5 conditions. The gap follows the three context units.

Condition	Context	Answer the pattern asks for
lit	A, B, A	B
ctl	A, B, C	none (lag control)
rel+k, $k \in \{+2, +5, -3\}$	A, B, T_kA	T_kB
seq	A, T_2A, T_4A	T_6A
ana	A, T_5A, B	T_5B
null	silence	none

S5 Motif development (E5, preregistered)

S5.1 Design

Songs return to earlier material, often changed: a phrase comes back, comes back higher, or continues a pattern. E5 asks which of these an audio song model does when it fills a gap. Each item is a context of three 4 s units (0 to 12 s), a gap from 12 to 16 s that the model fills by repaint, and a unit D after the gap. The units are short random melodies rendered as a solo instrumental line. Table 1 lists the eight conditions. In *lit* the pattern A B A asks for B. In *ctl*, A B C, nothing asks for B, but B sits exactly as far before the gap (8 s) as in *lit*, so the contrast separates a repeat driven by the pattern from a repeat of whatever was heard 8 s earlier. T_k transposes by k semitones.

Every possible answer is rendered exactly from symbols, together with a foil $\tilde{H}$ for each hypothesis H: a derangement with the same rhythm and the same multiset of pitches. The indices are differences of accuracy (Section S2.2) against an answer and against its foil, so 0 is chance. CI $= \mathrm{acc}(B) - \mathrm{acc}(\tilde{B})$ in lit, ctl or null. In rel+k, the relational index RI_k uses T_kB and the literal index LI_k uses the unmoved B. ANA uses T_5B in ana. H4 uses the share of seq gaps that a classifier with a threshold $\tau = 0.7$, calibrated on renders only, assigns to T_6A.

The study was registered on 2026-10-26 at 17:14:21Z (`prereg.json`) with these hypotheses, thresholds and stated priors:

- H1, content copy: CI in lit ≥ 0.25, with the 95% cluster-bootstrap interval above 0 (prior about 0.65).
- H2, copy driven by the pattern rather than the lag: CI(lit) − CI(ctl) ≥ 0.15, with the interval above 0 (prior about 0.45).
- H3, transposition: for at least 2 of the 3 values of k, $\mathrm{RI}_k \geq 0.15$ with

the interval above 0, RI_k minus RI_k under ctl ≥ 0.10, and $\mathrm{RI}_k > \mathrm{LI}_k$ (prior about 0.2).

- H4, sequence extension: in seq, T_6A in at least 20% of gaps, T_5A and T_7A each in at most 5%, and the T_6A rate above its rate under null (exact binomial $p < 0.01$) (prior about 0.1).
- H5, analogy: ANA ≥ 0.15 with the interval above 0 (prior about 0.15).

Kill criteria included K0 (the ruler must sort hypothesis renders at least 95% correctly and detect one wrong note with AUC ≥ 0.9, otherwise the units are redesigned before any model run), K1 (in a pilot phase at least 75% of gaps must pass a quality gate), K2 (an interim futility check on the first 6 melodies) and K3 (CI(ctl) $\geq$ CI(lit) -0.05 would refute H2). The registered paper-level rule says that if the quality gates fail, the run is an infill-quality failure and must not be reported as an induction finding. Two further hypotheses on which layers make the copy (H6) and on a causal pointer (H7) belong to later phases and were not run.

The quality gate passes a gap if its voiced fraction is at least 0.6, its Basic Pitch polyphony is at most 1.3 and its level is within 6 dB of the context.

S5.2 The first run failed its quality gate (v1)

The first confirmatory run (GPU run 1, 2026-10-26) failed. Only 59 of 120 non-null gaps passed the gate (49.2%, where 75% was required): one of the two seeds passed 59 of 84, and the other passed 0 of 36, giving quiet, mostly unvoiced gaps for every melody (`v1_outcome.json`). By the registered rule this run supports no claim about motif development, positive or negative. The registered diagnosis lists four causes. All 144 repaints had received ACE-Step's text-to-music instruction instead of its repaint instruction, because the Python interface we called does not substitute the task instruction as the other interfaces do. Within a seed every item started from the same noise, so the whole grid held only two independent noise draws, and the caption had been chosen on one of them. Under identical noise, the text conditioning alone switched a pilot gap between audible and near-silent, so conditioning and noise interact; which of the two dominates was not tested. And the source clips were 20.4 to 21.2 s long instead of the intended 18 s. Separately, the file records a deviation: the registered K1 rule named flute, synth voice and clarinet captions, while the pilot had compared three flute captions, and this was not disclosed before the run. The registered indices were computed anyway and are recorded, not interpreted (for example CI(lit) 0.159 [0.096, 0.232]).

S5.3 The corrected run (v2)

Addendum 2 (2026-10-27, 18:08:52Z) registered one retry before any of its generations. It kept H1 to H5 and every threshold, and changed the procedure: 36 new melodies; 4 draws per melody, each with its own hashed pair of seeds shared by all 8 conditions; the repaint instruction; deterministic kernels and a fixed decode chunk; and a pilot (Phase 0b) on the 14 v1 melodies (mel00 to mel13: the 2 v1 pilot melodies and the 12 v1 confirmatory melodies, used here as pilot material) that chooses among 12 cells (repaint or text-to-music instruction, source as rendered or cut at 18 s, and flute, synth voice or clarinet). The primary population is ALL draws, scored with the registered estimator; the gate-passing draws (PASS) are secondary, and every clause that compares two conditions must also hold in PASS. The number of melodies and draws was chosen by a simulation calibrated on the v1 outcomes, which the addendum discloses.

Before any v2 model run, K0 failed, for a reason in the procedure. The registered wrong-note draw picks any note of B, but the ruler scores only notes inside the gap minus 0.4 s at each edge, and B's first note starts at 12.0 s. In the first set of v2 melodies 2 or 3 notes of each B were never scored, so a wrong note landed on an unscored note with probability 0.275, and even a perfect ruler would expect a one-wrong-note AUC near 0.86. The measured AUCs were 0.848 (flute), 0.861 (synth voice) and 0.806 (clarinet), against 0.9 required. The registration prescribes a redesign of the units after a failed K0. The author chose that remedy over amending the check, and the choice with its exact procedure was committed and pushed (commit `1c91106`, recorded at 17:20:04Z) before K0 ran again, once, with no retry (a first start of that rerun was stopped by a tool time limit before it wrote any file, and was started again unchanged; addendum 2 records this). In the rebuilt melodies every note of A, B, C and B2 lies inside the scored window. K0 then passed in all three timbres: classification 0.983, 0.987 and 0.985, one-wrong-note AUC 0.965, 0.996 and 0.944, two-wrong-note AUC 1.0, 1.0 and 0.999 (flute, synth voice, clarinet; `results/gpu_run2/e5v2_setup.json`).

Run A ran on an NVIDIA L4 on 2026-10-28 at commit `83d46e7`. Phase 0b made 1344 repaints, 112 gaps per cell. Three cells passed K1 (84 of 112 needed): repaint with the source as rendered, for flute (88), clarinet (88) and synth voice (84). The other nine passed between 10 and 65. The registered tie order chose flute (`e5v2_phase0b.json`). The confirmatory grid then ran 36 melodies $\times$ 8 conditions $\times$ 4 draws = 1152 repaints, with none missing or lost.

Table 2: E5 v2 registered indices, ALL draws (36 melodies, 144 draws per condition), mean and 95% cluster-bootstrap interval over melodies (10,000 resamples). Source: `results/gpu_run2/e5v2_main.json`.

	Index	Value [95% CI]	Threshold	Verdict
H1	CI(lit)	0.471 [0.405, 0.536]	≥ 0.25	supported
H2	CI(lit) − CI(ctl)	0.365 [0.306, 0.425]	≥ 0.15	supported
H3	RI_{+2}	−0.021 [−0.043, 0.001]	≥ 0.15	not met
	RI_{+5}	0.002 [−0.024, 0.028]	≥ 0.15	not met
	RI_{-3}	0.020 [−0.002, 0.041]	≥ 0.15	not met
H4	T_6A in seq	0 of 144	$\geq 20\%$	not supported
H5	ANA	0.008 [−0.018, 0.034]	≥ 0.15	not supported
	CI(ctl)	0.106 [0.063, 0.152]		
	LI_{+2} / LI_{+5} / LI_{-3}	0.131 / 0.098 / 0.103		
	CI(null)	−0.025 [−0.044, −0.007]		

S5.4 Results

Quality passed: 804 of 1008 non-null gaps passed the gate (0.798; 756 needed), and 246 of 252 melody-condition cells had at least 2 passing draws (189 needed). K2 and K3 were not triggered: on the first 6 melodies CI(lit) was 0.520 and CI(ctl) 0.077, where K2 needs both below 0.10. The registered paper-level outcome is `supported:H1,H2` (`results/gpu_run2/e5_terminal.json`). Table 2 gives the indices and Figure 5 shows them; Appendix SB has every index in both populations.

H1 and H2, supported. After A B A the gap matches B with CI 0.471 [0.405, 0.536]; in absolute terms it matches 0.66 [0.60, 0.71] of B's notes, against 0.07 in null gaps. After A B C it matches B far less, 0.106 [0.063, 0.152], and the difference, 0.365 [0.306, 0.425], clears the H2 threshold. It also holds in PASS: 0.403 [0.327, 0.479]. The classifier assigned 74 of 144 lit gaps to B and 12 of 144 ctl gaps. One check matters for H2: lit gaps passed the quality gate more often than ctl gaps (0.875 against 0.778; exact McNemar 17 against 3, $p = 0.0026$). The rate difference, 0.097, is under the 0.10 that would add the registered "differential gating" label, and the verdict uses all draws, so the gate does not select the H2 population. Exploratory Lee bounds, which trim the passing lit draws down to the ctl pass rate, put the PASS difference between 0.363 [0.289, 0.436] and 0.471 [0.407, 0.535].

H3, not supported for any k. After A B T_kA the gap does not move B by k: RI_k is −0.021, 0.002 and 0.020 for $k = +2$, $+5$ and -3, against

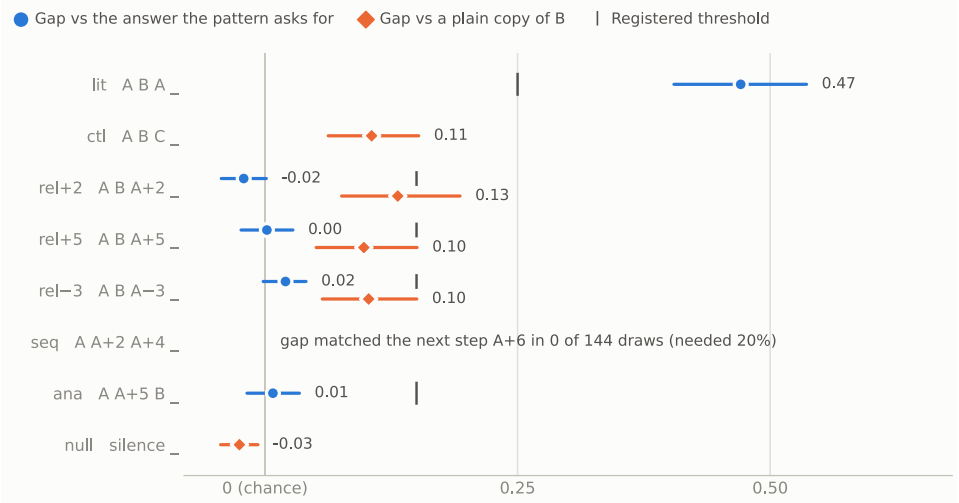


Figure 5: E5 v2: each index is the match to an answer minus the match to its foil (0 is chance), mean and 95% CI over 36 melodies, 4 draws per melody and condition. Blue: the answer the pattern asks for. Orange: an unmoved copy of B. Ticks: registered thresholds. Source: `results/gpu_run2/e5v2_main.json`.

0.15 required. The gap matches the unmoved B (LI_k 0.131, 0.098 and 0.103) at about the level of the lag copy in ctl, where B sits at the same distance before the gap and nothing asks for it (CI(ctl) 0.106 [0.063, 0.152]). $RI_k - LI_k$ is negative for every k (-0.152 [-0.221, -0.089], -0.096 [-0.158, -0.040], -0.082 [-0.136, -0.030]). Each LI_k is below 0.15, so the registered "literal-lag copier" label does not apply either. The largest relational effect, RI_{-3} compared with its value under ctl, is 0.028 [0.005, 0.050] in ALL and 0.029 [0.003, 0.054] in PASS. Its interval excludes 0, but the value is about 0.03 against the 0.10 that this clause of H3 requires. It is one of nine H3 clause comparisons and is not adjusted for multiplicity, so we treat it as exploratory.

H4, not supported. In seq the gap was never classified as the next step T_6A: 0 of 144 draws, and 0 of 144 under null and under ctl, so both registered tests give $p = 1$. The classifier chose T_2A, the unit that starts 8 s before the gap, in 19 of 144 draws (0.132), T_4A, the unit just before the gap, in 5 (0.035), D in 6 (0.042), and no hypothesis in 114 (0.792).

H5, not supported. After A T_5A B the gap does not match T_5B: ANA is 0.008 [−0.018, 0.034] (PASS 0.007 [−0.025, 0.038]), against 0.15 required.

Notes on the run. The null gaps were expected to be nearly silent and were not: their level had a median of −21.7 dBFS (range −46.4 to −12.8), and 21 of 144 were classified as D, the unit after the gap. This makes the null comparison of H4 easier to pass, and it made no difference here, because the T_6A count was 0 in every condition. The run state was kept on the notebook's local disk rather than on durable storage; the run went through in one pass with no interruption. After the run, the registered hashes, the ledgers (one pass, one attempt, no missing, extra or duplicate job) and the seed table were checked, and every index was recomputed with the repository's code and with separately written code; both reproduced the stored values (run log).

S5.5 What E5 shows

Given A B A, the gap matches B's notes at 0.66 on average and is classified as B in about half the draws (74 of 144), far more often than after A B C (12 of 144), so the repeat depends on the pattern and not only on B having been played 8 s earlier. The model does not carry a transposition over to B, extend a sequence or complete an analogy at the registered thresholds. Where it fails, the gap matches the unmoved B only at the level of the lag copy seen in ctl (LI_k 0.10 to 0.13 against CI(ctl) 0.11). These are pitch-tracker scores of instrumental flute lines filled by repaint, in one model with the 8-step sampler. The listening test of Section S8 will check whether people hear the same repetition.

S6 Lyric alignment heads and the lyric clock

S6.1 Do the lyric heads follow the words? (E6 read-out)

ACE-Step 1.5's code reads lyric timestamps from seven cross-attention heads of its renderer: layer 2 head 6, layer 3 heads 10 and 11, layer 4 head 3, layer 5 heads 8 and 9, and layer 6 head 8 (the defaults at commit `ca1e85f`). Before building on them, the E6 read-out asked whether they give the onset of each sung word. It was registered in addendum 2 as exploratory, with a rule that decides only whether the clock stage B1 may run.

The read-out generated 88 a cappella text-to-music clips: the 12 E4 lines × 4 seeds with E4's caption, and 4 four-line verses × a female and a male

a cappella caption × 5 seeds. It followed ACE-Step's own timestamp path: the generated latent is noised again to $t = 1/8$ with seed 42 and passed once through the renderer, and an aligner with dynamic time warping maps the lyric tokens to the 80 ms frames from the attention of the seven heads. A word's onset is the start of its first token. The control is the same pass read through 200 random sets of 7 heads, matched by layer and sharing no key or value head with the alignment heads. The reference is Whisper large-v3's word time stamps. A clip is evaluable if it passes the vocal gate and at least 4 lyric words are matched to Whisper.

All conditions of the rule held (`results/gpu_run2/e6_gate.json`, `e6_readout.json`). All 88 clips were generated and scored. 76 passed the vocal gate (44 needed) and 50 were evaluable (20 needed); another 26 vocal clips had fewer than 4 matched words. The median over evaluable clips of the per-clip median onset error was 0.21 s (at most 0.25 s allowed). The alignment heads beat the median of the random sets in 49 of 50 clips (0.98; 75% needed); the random sets' median error was 1.95 s, and none of the 200 sets did as well as the alignment set. ACE-Step's own timestamp function worked on every clip, and our implementation reproduced it. So B1 was allowed to run.

Outside the rule, 95% cluster-bootstrap intervals were 0.795 to 0.932 for the vocal rate, 0.18 to 0.22 s for the error and 0.944 to 1.0 for the share of clips where the heads beat the random median. The heads' onsets lag Whisper's by a median of 0.20 s, 65% of matched words are within 0.25 s, and other re-noising seeds (43 and 44) give 0.20 and 0.21 s. Read at each of the 8 sampling steps instead of after generation, the error is 0.18 to 0.27 s. Two cautions limit what 0.21 s means. Whisper and a forced aligner (MMS; Pratap et al., 2024) differ by a median of 0.39 s on these clips, so the error is within the uncertainty of the reference. And six of the seven heads are the six best single heads (0.10 to 0.22 s), but the seventh, layer 3 head 11, is off by 2.41 s on its own. The read-out shows that the heads follow word timing in finished audio. It does not show that they control it. The ACE-Step 1.5 report describes a reinforcement-learning reward computed by dynamic time warping over lyric cross-attention maps [Gong et al., 2026]; it does not say which heads, but its training diagram ends the turbo model's training with a reinforcement-learning stage after the few-step distillation. If that stage used this reward, heads that follow word timing are expected, so beating random heads is a sanity check rather than a finding.

S6.2 The clock

The lyric clock adds a fixed value β to the pre-softmax cross-attention logits of chosen heads in layers 2 to 6, at all 8 steps. The bias applies from the query frames whose centre lies inside word j's window to the keys of word j's lyric tokens. Word j is sung on note j, and its window runs from that note's written onset to the next note's onset (the last word's to its note end). All other logits are untouched: lead-in frames, frames after the last note, the structure tag, the header, the caption keys and all other heads. If a head gives a word attention probability p inside its window, the bias raises it to $pe^{\beta}/(pe^{\beta} + 1 - p)$: from $p = 0.10$ to 0.45 at $\beta = 2$, 0.86 at $\beta = 4$ and 0.997 at $\beta = 8$. Head set A is the seven alignment heads; set W is all 80 heads of layers 2 to 6.

S6.3 B1: the clock alone (registered; not supported)

B1 was fixed in addendum 2 before Run A, and its code was registered in a separate code addendum (2026-10-28, 20:56:49Z) after a CPU test showed the hook bit-identical at $\beta = 0$. It tests timing alone, with text-to-music and no melody guide. The items are the 12 E4 lines with E4's caption, 3 draws each, and five arms share each draw's seeds: no hook; $\beta = 1$, 2 and 4 on set A; and $\beta = 2$ on a random set of 7 heads, a different registered set for each draw. That is 180 runs. A clip's timing error is the median distance between Whisper's word onsets and the written note onsets over its matched words; a line's error in an arm is the median over its evaluable draws, and its reduction is $1 - \text{error(arm)}/\text{error(no hook)}$. The rule, at $\beta = 2$: (1) a reduction of at least 30% in at least 9 of the 12 lines; (2) word recall and vocal-gate rate each no more than 0.10 below the no-hook arm; (3) a median reduction with random heads below half that with set A.

All 180 runs were generated and scored (`results/gpu_run3/b1_result.json`). Without a guide the model sings these lines early: in a typical clip the words start 3.62 s before their written notes. Condition 1 failed. At $\beta = 2$, 2 lines improved by 30% or more (0.41 and 0.77), the median reduction was 0.047, and the closest miss was 0.28. Conditions 2 and 3 held: recall went from 0.595 to 0.600 and the vocal-gate rate from 0.944 to 0.972, and the random heads' median reduction was -0.001, below half of 0.047. The registered verdict is *not supported*. The result file's label "timing not moved in 9 items" is a fixed phrase for a failed condition 1; in fact 10 of the 12 lines did not improve.

As dose-response outside the rule, $\beta = 1$ improved 1 line (median reduc-

tion 0.003) and $\beta = 4$ improved 6 (median 0.229, 95% interval over lines 0.008 to 0.365), with recall down to 0.525 and the words still starting a median 2.92 s early. A stronger clock moves timing somewhat; the registered strength does not.

S6.4 B2: the clock with a melody guide (registered)

Design. B2 combines the clock with the guides of E4. It was registered in addendum 3 (2026-10-29, 04:50:31Z) before any full-scale B2 generation. Its claim concerns a pipeline that a user could run: one recipe fixed on development lines, the user's melody rendered as a wordless guide, the user's lyric, 4 draws, and an automatic selector that keeps the draw with the largest q, which needs only the melody and the lyric. Stage D chose recipes on the 12 E4 lines with 3 draws each, from 30 configurations: guide Voice Oohs or clarinet, context held for $k = 3$, 4 or 5 steps, and clock none, A at $\beta = 2$, 4 or 8, or W at $\beta = 2$. A registered pure function chose S^*, the clocked configuration with the most lines whose best of 3 draws is joint, among those whose clips passed the vocal gate at least 60% of the time, and B^*, the best unclocked configuration by the same key. Stage T then ran on 24 new lines (11 to 13 one-syllable words, written to fixed rules and checked in code), new melodies from E4's generator, new seeds and 4 draws, with these arms: S^*; $\mathrm{S}^*_{\mathrm{b0}}$, the same guide and k without the hook; $\mathrm{S}^*_{\mathrm{rand}}$, S^* on a random set of 7 heads, a different registered set for each draw; $\mathrm{S}^*_{\mathrm{shuf}}$, S^* with the windows rotated by half a line so each word gets another word's note; the unclocked E4 settings with the Voice Oohs guide at $k = 3$ and $k = 4$; a no-guide reference; and B^*. Arms that coincide are run once.

J is the share of the 24 test melodies whose kept draw is joint. The rule has three parts. P0 is integrity: hashes, the selection fixed before Stage T, the A0 checks, 40 hook calls in every hooked run, one guide latent across k, the pinned Whisper revision, and at most 5% of runs missing. P1, clock credit, requires $\Delta J = J(\mathrm{S}^*) - \max_b J(b) \geq 0.20$ over the unclocked guided arms and a 2.5th percentile of the bootstrap ΔJ (10,000 resamples of melodies, the maximum retaken in each) above 0. P2, capability, requires $J(\mathrm{S}^*) \geq 0.50$, at least 12 of 24 melodies. The verdict is "B2 supported" if P0, P1 and P2 hold, and "Capability reached, clock not credited" if P0 and P2 hold and P1 fails. The registered power simulation, calibrated on E4, gave P1 a probability of 0.27 and 0.65 in its two smallest non-null effect cells (`experiments/e8_clock_sing/power_b2.json`).

The timing trim. Before Stage D, the A0 check projected the run's length from its own clips and, past a 7-hour budget, dropped configurations by a registered rule. It projected 32.6 h, so it dropped the five clarinet configurations at $k = 3$ (29.4 h) and then the no-guide arm of Stage T (29.1 h) (`a0_report.json`). The projection was far too high: the stages took 283 minutes in all (`status.json`). Three of A0's five guided clips were Voice Oohs covers on which Whisper looped, so the median scoring time of a guided clip was 49.9 s instead of a few seconds, and the registered allowance for loops was added on top. The run log estimates that the measured costs would have projected 6.2 to 6.6 h, under the budget. The trim followed the registered rule, but clarinet at $k = 3$ was never tested and Stage T has no no-guide reference. That arm was a reference for words and the vocal gate, not a comparator, so no verdict depends on it. Whether clarinet at $k = 3$ would have changed the selection cannot be known; the clarinet recipes that ran were joint on at most 3 of the 12 development lines.

Stage D and the selection. Stage D ran 900 clips (25 configurations × 12 lines × 3 draws), of which 690 passed the vocal gate, and 13 of the 20 clocked configurations were eligible (`b2_selection.json`; Appendix SB). S^* is the Voice Oohs guide for the first 3 of 8 steps with the clock at $\beta = 8$ on set A, joint on 9 of the 12 development lines; the next was the same at $k = 4$, with 6. B^* is the same guide at $k = 3$ without the clock, joint on 2. So ${S^*}_{b0}$, the E4 setting at $k = 3$ and B^* are one recipe, run once. The selection file was written and its hash put at the head of the Stage T ledger at 00:07:20Z on 2026-10-30, before the first Stage T generation.

Stage T and the verdict. All 480 runs (5 arms × 24 melodies × 4 draws) were generated and scored, and P0 held (`results/gpu_run3/b2_result.json`). Table 3 gives the arms.

P2 held: the kept draw of S^* was joint on 14 of 24 melodies, where 12 were needed. P1 failed. The best comparator was the same guide without the clock, with 8 of 24, so $\Delta J = 0.583 - 0.333 = 0.25$, above the 0.20 needed, but the bootstrap's 2.5th percentile was exactly 0, and the rule needs it above 0. The registered verdict is *Capability reached, clock not credited* (`b2_summary.json`).

The 2.5th percentile of 0 does not depend on the resampling seed or the implementation. 286 of the 10,000 resamples had $\Delta J \leq 0$ (2.9%) and 104 had $\Delta J < 0$ (1.0%); these shares are not a p-value that the rule uses. Recomputed from the clip rows, the kept draws and every arm's J match the

Table 3: B2 Stage T, 24 new melodies × 4 draws per arm. S^*: Voice Oohs guide for 3 of 8 steps, clock at $\beta = 8$ on the seven lyric heads. S^*_{b0}: the same guide without the clock (also B^*). S^*_{rand}: the clock on random heads. S^*_{shuf}: the clock with rotated windows. $k = 4$: the guide for 4 steps, no clock. J: melodies whose kept draw (best of 4) is joint. Joint: clips with $q \geq 0.5$, of 96. Kept q: mean q of the kept draws. NPA_v and WR: means over 96 clips (NPA_v chance about 0.10). CTC: melodies whose kept draw is joint when WR comes from the CTC recogniser. Onset: median word-start error (s) over clips with at least 4 matched words. Source: `results/gpu_run3/b2_result.json`.

Arm	J	Joint	Kept q	NPA_v	WR	CTC	Onset
S^*	14	31	0.517	0.44	0.50	2	0.42
S^*_{b0}	8	13	0.385	0.45	0.32	3	1.73
S^*_{rand}	8	11	0.353	0.44	0.28	1	1.86
S^*_{shuf}	7	9	0.391	0.41	0.31	2	4.49
$k = 4$, no clock	3	5	0.175	0.75	0.07	1	3.75

stored values, and the 2.5th percentile is 0 under resampling seeds 0 to 19 as well (2.8% to 3.5% of resamples at or below 0). Counted exactly over all 24^{24} equally likely resamples, the ideal bootstrap gives $P(\Delta J \leq 0) = 3.0\%$ and $P(\Delta J < 0) = 1.3\%$, so its 2.5th percentile is exactly 0 too. Melody by melody, S^* won 8, lost 2 and tied 14 against the same guide without the clock (exact sign test on the 10 untied melodies, two-sided $p = 0.11$). These checks were made after the run and are exploratory (`results/checks/runb_secondary.json`).

Labels and contrasts. The registered labels never change a verdict. Strict capability: no; S^* had $q \geq 0.75$ on 2 of 24 melodies, as did the guide alone. Head-specific clock: yes; mean kept q is 0.517 for S^*, 0.353 with random heads and 0.385 without the clock. Timing-specific: yes; the lower bound of the S^* minus S^*_{shuf} contrast in mean kept q is 0.049 (mean 0.126). Part of this contrast comes from the in-order matcher, which penalises words sung on other words' notes: with words counted in any order (a registered secondary measure), the rotated-window arm is joint on 10 of 24 melodies instead of 7 and S^* on 15 instead of 14, and the contrast falls to 0.078 [0.015, 0.141] (post-hoc and exploratory, with the registered item bootstrap; `results/checks/runb_secondary.json`). The contrast with the guide alone does not shrink (0.139 [0.068, 0.210]). Timbre copy: no; none of S^*'s 96 clips

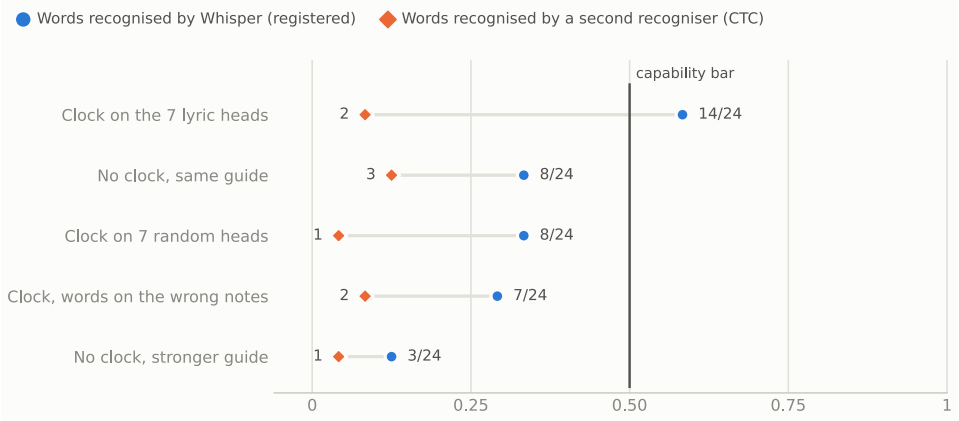


Figure 6: B2 Stage T: share of 24 new melodies whose kept draw (best of 4) has at least half the notes and half the words right. Blue: words by Whisper, as registered. Orange: words by the CTC recogniser on the same kept draws. The line marks the registered capability bar (P2). Source: `results/gpu_run3/b2_result.json`.

was vowels only. Words cut: no; recall is 0.503 in the first half of the line and 0.489 in the second. Frontier moved: no; NPA_v stayed level (-0.006) and WR rose by 0.174, short of the 0.20 required. Development futility: no. Independent recogniser: P2 fails with CTC word recall on the same kept draws, 2 of 24.

Outside the rule, the paired contrast of mean kept q between S^* and the guide alone is $+0.131$ (95% interval 0.064 to 0.200); against random heads $+0.164$ (0.092 to 0.240); against rotated windows $+0.126$ (0.049 to 0.205). The clock also moves timing under a guide: the median word-start error is 0.42 s with the clock, 1.73 s without it, and 4.49 s with rotated windows (Table 3). These medians come from different clips (71 and 46 clips with at least 4 matched words); on the 39 seed-paired clips that have an onset error in both arms, the medians are 0.40 s with the clock and 1.72 s without it (post-hoc and exploratory; `results/checks/runb_secondary.json`). The rotated windows are worse on timing by construction, since they move each word to another note, so that arm is a check that the manipulation works. Fewer S^* clips are wordless (0.021, against 0.281 for the guide alone), but the rotated-window clock removes wordless outputs almost as well (0.031). Part of the clock's effect on words therefore does not depend on its timing; the in-order word count and J favour the correct windows.

S6.5 What limits the B2 result

Several things limit what B2 shows.

The clock is not credited. The gain in J is 0.25, above the size bar, but its bootstrap interval reaches 0, and the rule was written so that this case does not credit the clock. The paired contrast of mean kept q excludes 0, but that score is not the registered criterion, it also rests on Whisper, and it uses the same best-of-4 selection. The test also had little power at this effect size. In the registered simulation, P1 passed with probability 0.27 in the cell whose mean ΔJ was about 0.21 and 0.65 in the cell at about 0.34 (`power_b2.json`); the observed 0.25 lies between them. At an effect of the observed size the registered test was probably more likely to fail than to pass, so a failed P1 says little about whether the clock has an effect.

The words rest on Whisper. The second recogniser does not confirm them (Figure 6). With CTC word recall on the same kept draws, S^* is joint on 2 of 24 melodies and the guide alone on 3. On the kept draws, CTC rates S^* slightly below the guide alone (mean difference in WR -0.080; 9 melodies up, 13 down, 2 tied); over all 96 seed-paired clips it rates S^* slightly above (mean CTC WR 0.179 against 0.160). It reads every arm poorly, and the two recognisers agree least on S^*'s clips (per-clip correlation of WR 0.53, against 0.61 to 0.79 for the other arms). Because the kept draw is chosen by Whisper, a second recogniser scoring the same draws is biased against the arm that gains most from selection. With the draws re-selected by CTC, S^* is joint on 4 of 24 melodies and the guide alone on 3. The CTC model was trained on read speech and was not checked on singing (Section S2). Whisper's own chance level for these lines, scored against the other lyrics, is 0.14 for S^* and 0.09 for the guide alone, so the gain in WR above chance is about 0.12 rather than 0.174. This does not show that the words are unclear, and it does not confirm that they are clear. Whisper received no lyric prompt, and none of S^*'s 31 joint transcripts contains one of the registered hallucination phrases. The CTC comparisons on the kept draws, the correlations, the CTC re-selection, the chance levels and the hallucination check were made after the run and are exploratory (`results/checks/runb_secondary.json`).

The kept draw is the best of 4 by the score that decides it. This is the registered pipeline, and the selector uses only the melody and the lyric. Still, with a single draw S^* is joint on 0.32 of melodies on average over draws,

and the guide alone on 0.14; with 2 and 3 draws the values are 0.47 and 0.53 against 0.23 and 0.29.

The joint bar does not tie words to notes. NPA_v and WR are scored separately, so a joint clip need not sing the recognised words on the right notes (Section S2). The word-start error and the listening test address this only in part.

The margin is small. Two of the 14 joint melodies have no note to spare: with one note fewer right on every clip, $J(\mathrm{S}^*)$ would be exactly 12 of 24, and with two fewer, 10. The words have more room: one or two matched words fewer leave 14 and 12 (post-hoc; `results/checks/runb_secondary.json`). At the strict bar both recipes reach 2 of 24.

The trim removed part of the design. Clarinet at $k = 3$ was never tried in Stage D, and Stage T has no no-guide reference, because the A0 projection was about six times too long.

S6.6 Run C: B2's recipes on 96 new melodies (registered)

Design. B2's 24 melodies were too few to settle the clock's effect. Run C tested the two B2 recipes again, fixed in advance with no selection stage, on 96 new lines written to B2's rules and sharing no line with E4, E6, B2 or E7, with new melodies from B2's generator and new seeds. Its single stage, C1, ran the four arms of B2's Stage T that rest on S^* (S^*, $\mathrm{S}^*_{\mathrm{b0}}$, $\mathrm{S}^*_{\mathrm{rand}}$ and $\mathrm{S}^*_{\mathrm{shuf}}$) with 4 draws each, 1536 runs, scored by B2's scorer imported unchanged. There was no timing trim. The rule has P0 as in B2 and two parts. R1, the clock effect, holds if the bootstrap 2.5th percentile of $\Delta J = J(\mathrm{S}^*) - J(\mathrm{S}^*_{\mathrm{b0}})$ is above 0 (10,000 resamples of melodies). R2, the capability, holds if $J(\mathrm{S}^*) \geq 0.50$, at least 48 of 96 melodies. The verdict is "Clock helps, capability not reached" if P0 and R1 hold and R2 fails; the other verdicts are "Replicated", "Capability holds, clock effect not shown", "Not replicated" and, if P0 fails, "Technical". B2's 0.20 size bar became a label rather than a condition of R1, a change the addendum states before any Run C generation. R1 is therefore looser than B2's P1, with no size bar and a single comparator, so a pass credits a smaller effect than B2's rule asked for. Simulated from B2's per-melody outcomes, R1 would hold with probability 0.979 at B2's observed effect, 0.603 with the effect halved and 0.024 with no effect. R2 would hold with probability 0.965 at B2's observed values and 0.252 if S^*'s J fell to about 0.46 with the effect halved (`experiments/`

Table 4: Run C (C1), 96 new melodies × 4 draws per arm. Arms as in Table 3: S^*, the Voice Oohs guide for 3 of 8 steps with the clock at $\beta = 8$ on the seven lyric heads; S^*_{b0}, the same guide without the clock; S^*_{rand}, the clock on random heads; S^*_{shuf}, the clock with rotated windows. J: melodies whose kept draw (best of 4) is joint. Joint: clips with $q \geq 0.5$, of 384. Kept q: mean q of the kept draws. NPA_v and WR: means over 384 clips (NPA_v chance about 0.10). Wordless: share of clips that the registered scorer calls wordless. CTC: melodies whose kept draw is joint when WR comes from the CTC recogniser, on the mix and on the separated vocal. Onset: median word-start error (s) over clips with at least 4 matched words. Source: `results/gpu_run4/c1_result.json`.

Arm	J	Joint	Kept q	NPA_v	WR	Wordless	CTC mix	CTC vocal	Onset
S^*	47	83	0.465	0.43	0.44	0.036	9	9	0.40
S^*_{b0}	17	23	0.281	0.41	0.23	0.414	9	8	2.17
S^*_{rand}	20	25	0.286	0.41	0.19	0.419	3	6	2.02
S^*_{shuf}	9	10	0.312	0.37	0.26	0.089	1	2	4.79

`e9_clock_replicate/power_c1.json`). The figures at the observed values are optimistic, since they take as true the estimates from one small test of a recipe chosen on development lines. A new secondary measure applies the CTC recogniser to the separated vocal as well as to the mix. Run C was registered in addendum 4 (2026-10-30 08:57:04 UTC, commit `0c97266`) before any Run C generation, and a blind A/B pack of 24 further pairs was made the same way as B2's (Section S8).

The run. Run C ran in Colab on an NVIDIA L4 at commit `0c97266`. A0 passed at 09:10 on 2026-10-30, and the 1536 generations and scores ran from 09:10 to 14:33 (322 minutes). Every run succeeded on its first attempt, and none is missing or unscored. Two later sessions ran only setup, decide and package: they generated and scored nothing, reused A0 on the same runtime fingerprint and printed the same verdict (Appendix SA). P0 held (`results/gpu_run4/c1_result.json`).

The verdict. Table 4 gives the arms. R1 held: $\Delta J = 0.490 - 0.177 = 0.3125$, with a 95% bootstrap interval of 0.198 to 0.427, and none of the 10,000 resamples was at or below 0. R2 failed: the kept draw of S^* was joint on 47 of 96 melodies, where 48 were needed. The registered verdict is *Clock helps, capability not reached* (`results/gpu_run4/c1_summary.json`).

Labels and contrasts. The registered labels never change a verdict. B2's size bar: held; ΔJ is 0.3125 against the 0.20 that B2's P1 asked for, though the interval's lower end, 0.198, is just under 0.20. Strict capability: no; S^* had $q \geq 0.75$ on 8 of 96 melodies. Head-specific clock: yes; mean kept q is 0.465 for S^*, 0.286 with random heads and 0.281 without the clock. Timing-specific: yes; the lower bound of the S^* minus S^*_{shuf} contrast in mean kept q is 0.122 (mean 0.154). Timbre copy: no; 2 of S^*'s 384 clips were vowels only. Words cut: no; recall is 0.454 in the first half of the line and 0.435 in the second. Frontier moved: yes, unlike in B2; NPA_v stayed level (0.429 against 0.408), and WR rose by 0.217, just past the 0.20 required. Independent recogniser: R2 fails with CTC word recall on the same kept draws, 9 of 96 melodies on the mix and 9 on the separated vocal.

Outside the rule, the paired contrast of mean kept q is +0.184 against the guide alone (95% interval 0.140 to 0.229), +0.179 against random heads (0.136 to 0.223) and +0.154 against rotated windows (0.122 to 0.185). Melody by melody, S^* won 36, lost 6 and tied 54 against the guide alone (exact sign test on the 42 untied melodies, two-sided $p = 2.8 \times 10^{-6}$; post-hoc, `results/checks/runc_secondary.json`). The median word-start error is 0.40 s with the clock, 2.17 s without it and 4.79 s with rotated windows; these medians come from different clips (269, 125 and 148 clips with at least 4 matched words). As in B2, the rotated-window clock removes most wordless outputs (0.089 of clips, against 0.036 with the clock and 0.414 without it), yet it is joint on fewer melodies than the guide alone (9 against 17). Part of the clock's effect on words therefore does not depend on its timing, while the in-order word count and J favour the correct windows. Figure 7 sets the four arms against the capability bar.

The run was rechecked from its files with separately written code, written from addendum 4's text (run log). The 63 files that addendum 4 registers, and the 110 file entries of the earlier addenda, hash to their registered values at `0c97266`. Every planned run appears once in the ledgers, in the registered order, with the registered recipe, seeds, hook calls and guide latent. This recompute matched the verdict, every rule number and every label, and six other defensible readings of the scoring each left S^* at 47 melodies.

S6.7 What limits the Run C result

The words rest on Whisper. The second recogniser does not confirm the words (Figure 7). With CTC word recall on the same kept draws, S^* and the guide alone tie: 9 melodies each on the mix, and 9 against 8 on the separated vocal. CTC reads every arm poorly (mean WR on the vocal 0.14

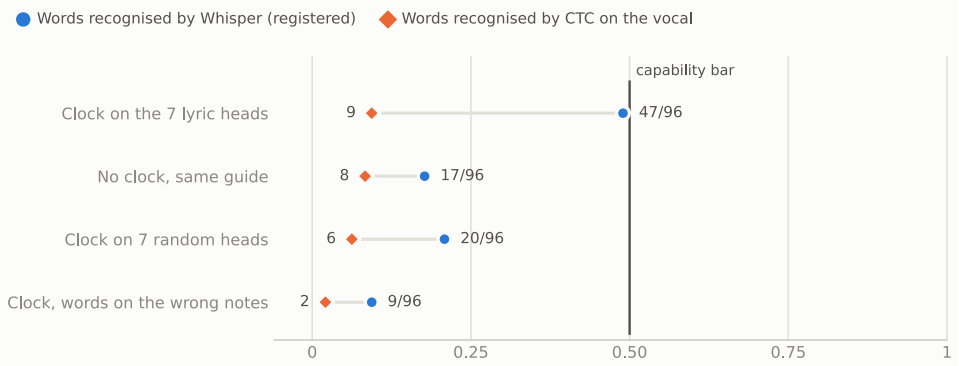


Figure 7: Run C (C1): share of 96 new melodies whose kept draw (best of 4) has at least half the notes and half the words right. Blue: words by Whisper, as registered. Orange: words by the CTC recogniser on the separated vocal, on the same kept draws. The line marks the registered capability bar (R2). Source: `results/gpu_run4/c1_result.json`.

with the clock and 0.12 without, against 0.44 and 0.23 for Whisper). Of the 47 melodies whose kept S^* draw is joint by Whisper, 9 (19%) are also joint by CTC on the vocal, against 8 of 17 for the guide alone, and in no arm is a kept draw joint by CTC but not by Whisper. Much of Whisper's gain is the clock turning wordless output into words: 41% of the guide alone's clips are wordless, against 4% with the clock. That flag is the registered scorer's, and it calls a transcript wordless if it is empty, at least half vowel sounds, or nothing but the registered hallucination strings and vowel sounds. The caveats given for B2 apply unchanged: the kept draw is chosen by Whisper, which biases a second recogniser scoring the same draws against the arm that gains most from selection, and the CTC model was trained on read speech and was not checked on singing. The CTC overlap and the wordless comparison were made after the run and are exploratory (`results/checks/runc_secondary.json`).

The capability bar sits inside the noise. R2 was missed by one melody. Over the registered resamples, $J(S^*)$ has a 95% interval of 0.39 to 0.59, and 46.1% of the resamples reach 0.50. With one correct note more or fewer on every S^* clip, 54 or 35 melodies would be joint, and with one matched word more or fewer, 54 or 40 (post-hoc; `results/checks/runc_secondary.json`). The miss does not show that the recipe falls short of half the melodies, and a pass by one would not have shown that it reaches them. The registered

verdict stands as the rule gives it.

Both recipes scored lower than in B2. $J(\mathrm{S}^*)$ fell from 0.583 in B2 to 0.490, a change of −0.094 (95% interval −0.31 to 0.13, with the two runs' melodies resampled independently), and $J(\mathrm{S}^*_{\mathrm{b0}})$ fell from 0.333 to 0.177, a change of −0.156 (−0.35 to 0.04). Neither drop is clear on its own. The registered power for R2 took B2's values as true (0.965). The observed $J(\mathrm{S}^*)$, 0.490, lies between B2's 0.583 and the 0.46 of the simulation's cell with S^* lowered, in which R2 held with probability 0.252.

The effect is not shown to be larger than B2's. ΔJ is 0.3125 in Run C and 0.25 in B2. The difference, 0.0625, has a 95% interval of −0.20 to 0.32, and 33.9% of the resamples are at or below 0; in mean kept q the difference is 0.053 (−0.03 to 0.14). In the point estimates, the whole widening comes from the guide alone falling further than S^* (by 0.156 against 0.094). The two runs' gaps cannot be told apart. Run C shows that the clock helps by the registered measure, not that it helps more than B2 suggested.

The pooled estimate is exploratory. Addendum 4 lists an estimate pooled over B2's 24 melodies and Run C's 96 as exploratory. It gives $\Delta J = 0.30$ (95% interval 0.19 to 0.40) and a mean kept q difference of 0.174 (0.135 to 0.211) (`c1_result.json`). Four fifths of its melodies are Run C's, so it adds little to Run C alone, and it changes neither run's registered verdict: B2's stays *Capability reached, clock not credited.*

The kept draw is the best of 4 by the score that decides it. With a single draw, S^* is joint on 0.216 of melodies on average over draws and the guide alone on 0.060, a gap of 0.156 (95% interval 0.10 to 0.21); in B2 the single-draw gap was 0.188 (0.07 to 0.30). With 2 and 3 draws the values are 0.34 and 0.42 against 0.11 and 0.15. Selection raises both recipes, and the gap is there without it.

The joint bar does not tie words to notes. As in B2, NPA_v and WR are scored separately, so a joint clip need not sing the recognised words on the right notes. The word-start errors and the blind A/B test address this only in part.

S6.8 What B1, B2 and Run C show

The lyric heads track word timing in finished audio. Biasing them at $\beta = 2$ without a guide did not move timing enough to pass B1, and a stronger bias moved it partly. With a wordless melody guide held for 3 of 8 steps and a bias of $\beta = 8$ on the same heads, the best of 4 draws passes the joint bar by Whisper on 14 of 24 new melodies in B2 and on 47 of 96 in Run C, against 8 of 24 and 17 of 96 for the same guide without the clock. B2's rule did not credit that gain. Run C's rule, looser and applied to four times as many melodies, does, and its controls (four random head sets, and windows moved to other notes) point to the lyric heads and to the note timing. With the clock the words start closer to their notes: a median of 0.40 s against 1.72 s on B2's 39 seed-paired clips, and 0.40 s against 2.17 s in Run C, each over the clips with at least 4 matched words (269 and 125). The capability bar was reached in B2 by two melodies and missed by one in Run C; in both runs, one or two notes more or fewer per clip would move the count across it. In both runs the second recogniser rates the clocked recipe and the guide alone alike, so the word gain rests on Whisper, and no listening result is in yet. Blind A/B tests of the B2 and Run C pairs are under way (Section S8).

S7 Discussion and limitations

Where the melody goes. In ACE-Step 1.5 a melody that has to pass through the 5 Hz tokens arrives in the output mostly lost, even when the tokens are computed from the melody's own audio. Rendered from those tokens, the output keeps 28% of the melody information that the input carries; the tokens change about as much with the lead instrument as with the melody; and a linear probe reads pitch from them far worse than from audio latents at the same rate, though well above what the token path renders. So part of the loss is in the tokenizer and part in how the renderer uses the tokens. The renderer passes a melody through when it receives the dense latents for all 8 steps, a full-strength cover close to a reconstruction. For a user who brings a melody, this places control in the renderer or in a better plan, not in the caption. We did not test the planner generating tokens from a symbolic melody, nor optimised tokens; E1 measures tokens that the model's own tokenizer makes from the audio, from one encoding and one seed, and does not bound what a better plan could carry.

Notes and words. With a wordless guide, the renderer trades words for notes (E4). The lyric clock was designed on the hypothesis that the trade

comes from a timing conflict between the guide and the model's own lyric timing. A simpler rival also fits E4: holding a wordless input longer keeps the output closer to it, words lost included. B2 and Run C give the same mixed picture. Under the clock the sung words start much closer to their notes (median 0.40 s against 1.72 s on B2's 39 seed-paired clips, and 0.40 s against 2.17 s over Run C's clips), and the in-order word count and J favour the correct windows over rotated ones (in Run C, 47 melodies against 9). But the rotated-window clock removes wordless outputs almost as well as the real one (0.031 against 0.021 in B2 and 0.089 against 0.036 in Run C, with 0.281 and 0.414 without the clock), so part of the clock's effect on words does not depend on its timing. B2's registered test did not credit the clock; Run C's, on 96 melodies and with a looser rule, does. That measure counts words by Whisper alone, much of its gain is the clock turning wordless output into words that Whisper finds, and by the CTC recogniser's count the clocked recipe and the guide alone come out about level in both runs (2 against 3 melodies in B2, 9 against 8 on Run C's separated vocal). Run C's gain cannot be told apart from B2's, and both recipes scored lower than in B2.

Motifs. The model repeats a phrase from the context when the pattern asks for it, in about half the draws by the classifier. When the pattern asks for a transposed phrase, the gap matches the unmoved phrase only at about the level at which, in the control condition, it matches the phrase heard at the same distance before the gap. It shows no sign of continuing a sequence. This matches the low priors registered for H3 to H5. For a user who relies on repaint, this suggests that a phrase can be brought back but not developed, with the caveat that the test used short instrumental phrases only.

Limitations.

- *One model and one sampler.* Every result concerns ACE-Step 1.5 with the 8-step turbo checkpoint (Runs A to C at commit `ca1e85f` with pinned weights; the first GPU run recorded neither the commit nor the weights revision). Other song models, checkpoints that sample with more steps, and other samplers may behave differently.

- *Synthetic melodies.* All melodies come from random generators and are rendered by FluidSynth, with no real singers and no real songs. The motif units are short instrumental lines; the B2 and Run C guides are General MIDI Voice Oohs.

- *One syllable per note.* E4, B1, B2 and Run C use English lines of 11 to 13 one-syllable words, one note per word, at 100 BPM, with one female a cappella caption. Melismas, multi-syllable words, other languages and accompaniment were not tested.
- *Words rest on Whisper.* Word recall is Whisper large-v3's. The CTC recogniser disagrees on B2 and on Run C, where only 9 of the 47 melodies joint by Whisper with the clock are also joint by CTC on the separated vocal, and no human has yet judged whether the words are clear.
- *Pitch rests on trackers.* Notes are scored by SwiftF0 on renders or separated stems. E0 shows what the trackers miss in a full mix; singing quality, timbre and naturalness are not measured at all.
- *Small samples.* E5 has 36 melodies, B1 12 lines, B2 24 test melodies and Run C 96. B2's verdict sits on the bootstrap boundary, its capability margin is two melodies, and at the observed effect its clock test had a power between 0.27 and 0.65. Run C missed its capability bar by one melody, a margin that one note more or fewer per clip would reverse, and its gain cannot be told apart from B2's (difference 0.06, 95% interval -0.20 to 0.32).
- *Selection and scoring.* In B2 and Run C the kept draw is the best of 4 by the score that decides it, and the joint score does not check that the recognised words are sung on the right notes. With a single draw, the clock's gain in J is 0.19 in B2 and 0.16 in Run C.
- *Deviations.* The first E5 run failed its gate; the E5 v2 melodies were rebuilt after a failed measurement check; the B2 timing trim removed one configuration family and one reference arm; two sessions reopened Run C's state after it had finished, and its result package was re-zipped by hand before it was checked, with no stored hash of the ledgers. Each is recorded (Appendix SA).

What would change these conclusions. Run C settled the question B2 left open under its own, looser rule: the clock helps by the registered measure. It did not settle the capability claim: its count fell one melody short of a bar that lies inside the noise of the measure, so whether this recipe reaches half of new melodies remains open. Listeners who do not hear the words that Whisper counts, or who do not hear the repeated phrase that

the tracker scores, would remove the corresponding claims; the blind tests of the B2 and Run C pairs ask whether people hear the melody and the words better with the clock. A different speech recogniser that agrees with Whisper on the clocked clips would strengthen the word result; one that agrees with the CTC model would weaken it further. For E1, a planner that writes better tokens from a symbolic melody, or a tokenizer with a higher frame rate or a pitch-specific code, would change where the melody is lost.

S8 Work under way

This section describes registered work whose results are not in this version: blind listening tests, whose ratings are open until 2026-11-30, and a comparison with a commercial song model, which has not been run. Their results will be added in a later version.

S8.1 Blind listening of the Run A, B2 and Run C packs

The final claims about how the outputs sound are meant to come from people listening blind. Three packs exist. The Run A pack holds 60 clips: 5 from each of the 8 E5 conditions and 20 E6 clips, renamed in an order drawn from the operating system's random source, with yes, no or unsure questions on whether vocals are present, whether words are intelligible, whether there is accompaniment, and whether the gap repeats an earlier phrase. The B2 pack holds the 24 Stage T pairs, each melody's kept S^* draw against the kept draw of the same guide without the clock, with the side of each pair drawn by the operating system's random source and written only to a separate key. Raters answer which clip has the melody right and which has the words clear (A, B, same or neither). The keys are archived privately and are opened once, after ratings close.

The analysis plan and its code were registered before any rating, at 2026-10-30 06:44:14Z in commit `83e8137` (`experiments/listening/registration/listening_plan.json`, SHA-256 `28c77329a8594cf25c1d39f75188042612cc9cda5ef965e4006888a00a1dd493`). For the B2 pack it computes, per question, a vote margin for the clock and tests it with a randomization test over the side assignments that put the clock on side A as often as the real key does; the margin is centred so that a listener who simply prefers the first or the second clip cannot move it. For the Run A pack it reports yes rates by condition, an exact permutation test of "the gap repeats an earlier phrase" between lit and ctl clips (with and without the author's own sheet), and agreement of the listeners' majority with the tracker's per-draw choice

and, for E6 clips, with the vocal gate and Whisper. Anyone may rate who has not heard the labelled demo clips, seen a key, read the per-clip results or opened the first, seed-ordered Run A pack, and no one who has seen a key decides when ratings close. Ratings close at the latest on 2026-11-30 at 23:59 UTC; exclusions and the hashes of all sheets are committed before a key is opened, and each key's salted fingerprint is in the plan.

The Run C pack holds 24 further pairs, each a Run C melody's kept S^* draw against the kept S^*_{b0} draw (Section S6.6). Its melodies and their order come from a seed registered in addendum 4, and the side of each pair was drawn by the operating system's random source and written only to a separate key. It shares no melody with the B2 pack and asks the same two questions. Its analysis plan applies the B2 pack's test and report unchanged, with the machine scores taken from `results/gpu_run4/c1_result.json`. A rater of this pack must also not have heard the labelled demo clips of the Run C package or read `results/gpu_run4/`, whose per-clip transcripts, read together with the demo list, tell which clip of a pair has the clock. Its ratings close on the same terms. All three packs are exploratory under addenda 2 to 4 and cannot change a registered verdict.

The Run C pack's plan was registered at 2026-10-31 06:58:28 UTC, before any rating, in commit `79e2868` (`experiments/listening/registration/listening_plan_c1.json`, SHA-256 `671badc5fc4931b73358c0390b47572418dafc36ee2fc94771449bfd387390fd`).

When this version was written, ratings were still open and no key had been opened. The number of raters and exclusions, the vote margins and tests for the B2 and Run C packs, and the Run A yes rates, the lit against ctl test and the agreement figures will be reported in a later version.

S8.2 Comparison with a commercial song model (E7 Stage S)

Stage S asks how well a commercial song model and ACE-Step 1.5 sing a melody that the user supplies. It was registered on 2026-10-27 at 19:49:06Z, before any song was generated. It uses 12 new eight-bar melodies (24 to 40 notes, 90 to 110 BPM) with original English lyrics, one syllable per note. The commercial model receives each melody through up to three input routes (a MIDI file, a piano recording and a wordless "oohs" recording), and ACE-Step 1.5 sings the same melodies with four settings from E4. The same tools score everything. Because commercial outputs add intros, repeats and small changes of tempo or key, the scorer first searches for the start, tempo and key, and every score is also computed against the 11 other melodies to remove

lucky matches. The registered decision rule says the commercial model already sings given melodies accurately if, on any route with at least ten complete items, its median note accuracy is at least 0.90 and its median word recall at least 0.60; otherwise the next step is a melody adapter trained inside the renderer. When this version was written, no song had been generated with the commercial model, and the comparison is left for later. If it is run, its scores will be published only if the model's terms of use allow it.

SA Preregistration and custody

SA.1 Registration files and time stamps

All registration files are in `experiments/e5_motif/registration/` of the public repository unless stated otherwise; the Stage S registration is in `experiments/e7_melody_adapter/` and the two listening plans in `experiments/listening/registration/`. A registration is not edited after it is committed, and changes are new files, with one exception: in the move to the public repository, `v1_outcome.json` had its path references (`refers_to` and `run`) updated to the new layout. Addendum 2 records this and the development file's SHA-256 (`1e5dc424eb4c8eca9e992bc8e9486cd969c79802eb3873dea85d30d25a255461`); the hash below is that of the public file. The time in the second column is the one written inside each file; the commit times are those of the public repository, where the push to GitHub is the external time stamp.

Table 5: Registrations, in order.

File	Written (UTC)	Commit and what it fixes
`prereg.json`	2026-10-26 17:14:21	private `6e459d8`; E5 H1–H7, K0–K5
`addendum_1.json`	2026-10-26 19:42:09	private `526fd3c`; package rename
`v1_outcome.json`	2026-10-26 21:50:00	E5 v1 gate failure and causes
`addendum_2.json`, step 1	2026-10-27 17:20:04	`1c91106`; K0 remedy R2
`addendum_2.json`	2026-10-27 18:08:52	`b6ed62f`; E5 v2, E6 read-out, B1
Stage S registration (E7)	2026-10-27 19:49:06	`f40132d`; commercial comparison
`addendum_2_b1code.json`	2026-10-28 20:56:49	`680b24b`; B1 code
`addendum_3.json`	2026-10-29 04:50:31	`bd61a9f`; B2
Listening plan	2026-10-30 06:44:14	`83e8137`; blind listening analysis of the B2 and Run A packs
`addendum_4.json`	2026-10-30 08:57:04	`0c97266`; Run C (C1)
Listening plan (Run C)	2026-10-31 06:58:28	`79e2868`; blind listening analysis of the Run C pack

The original registration and addendum 1 were committed in the author's private development repository before the public repository existed; addendum 2 records those commits. The public repository began as one commit of the restructured code, and addendum 2 records the file moves and that the moved code was unchanged apart from imports. Commit times: `1c91106` at 17:29:07, `b6ed62f` at 18:14:16 and `f40132d` at 19:51:09 on 2026-10-27; `680b24b` at 04:50:09 and `bd61a9f` at 04:52:49 on 2026-10-29 (UTC). The B1 code addendum was written on 2026-10-28 and committed with its code on 2026-10-29, before addendum 3. Run A started at 09:43:35 on 2026-10-28 at commit `83d46e7` (committed 09:33:17); Run B started at 21:01:47 on 2026-10-29 at commit `bd61a9f`. Addendum 4 and the Run C code were committed as `0c97266` at 09:01:03 on 2026-10-30, and Run C's setup began at 09:07 the same day at that commit; its A0 report is time-stamped 09:10:32. The Run C pack's listening plan was committed as `79e2868` at 06:59:04 on 2026-10-31.

SHA-256 of the registration files:

- `prereg.json`:
  `d162a0fdb0837d37a109f01c65de1ce1dfa52ba37ec0a7850d3955eed70d29bd`
- `addendum_1.json`:
  `ae541a4c0cc166e5d295fb91e18cbbdb218f28092ad5edb69d5ce55a7e8bdadb`
- `v1_outcome.json`:
  `bc4f36f6b715c92a2d7d7af878a4bde00579234ca1f3a6813a664caeffe66424`
- `addendum_2.json`:
  `81f49fae47d58701328227b6a768c9433cc4ae23a715edd2dd67f482c6f54144`
- `e5v2_seeds.json`:
  `72563da0f072f3ec631d6c5449d887b075e097a8d13f0bcdfecd3f84ff93b7b5`
- `addendum_2_b1code.json`:
  `e3f74a031352dda9aa77bd2e661ff2456dc42bbdd783e4be2c108b27303e1f04`
- `addendum_3.json`:
  `52844b55e45ae0bdaf1fb07d72d6038364956f8ede0feff0d2c9e95246bedd19`
- Stage S registration (E7):
  `95d01d3c0f24755ed6b27a89082746dd8095f7fd1e67d2ea79705d4a3cd8c1fd`
- `experiments/listening/registration/listening_plan.json`:
  `28c77329a8594cf25c1d39f75188042612cc9cda5ef965e4006888a00a1dd493`

- `addendum_4.json`:
  `e2b06c1ed7252ccb46a53f9315d5726d2eaac835ef7f04c06738fa77ca0029e4`
- `experiments/listening/registration/listening_plan_c1.json`:
  `671badc5fc4931b73358c0390b47572418dafc36ee2fc94771449bfd387390fd`

Each addendum lists the SHA-256 of every code, data and test file it registers, and the runs check them before they generate and again before they decide. Run A checked 18 code and 14 data hashes (`e5v2_main.json`); after Run B, all 89 files registered in the three addenda were found to hash to their registered values at `bd61a9f` (run log). After Run C, the 63 files that addendum 4 registers and the 110 file entries of the three earlier addenda hashed to their registered values at `0c97266` (run log).

SHA-256 of the main result files:

- `results/gpu_run2/e5v2_main.json`:
  `acc176047787b54559a62ebf8bb47bd82882ef54d5cd381986837219ad880e5d` (also stored in `e5_terminal.json`)
- `results/gpu_run3/b2_selection.json`:
  `047c73efcbd140bcb25ee66b5083018f237a7e28da9a7141707fc90839018170` (also the first line of the Stage T ledger)
- `results/gpu_run3/b2_result.json`:
  `913ac5fdb1b8113cd5f557e1e0970cd2e51ed671de78b99f3bbe6ae8264fe037`
- `results/gpu_run3/b1_result.json`:
  `bef42d535dc55fbfdd114a276b8eabc4547fd1e7af037cf8202be3339a3a19b9`
- `results/gpu_run4/c1_result.json`:
  `547b40c44fb2e739d551aa206e6fd96aaf1b68aa29e0b3a643032ad77a94aae3`

SA.2 Earlier looks at data

Each registration lists what had been run before it. Before `prereg.json`: one CPU repaint of pilot melody mel00, which is excluded from all confirmatory tests, after which a third pilot caption was added. Before addendum 2: every v1 outcome, a four-run CPU pilot on mel00, an outcome-free run of the quality gate on exact renders, one CPU clip of the E6 read-out with Whisper small.en, and an end-to-end CPU test of the read-out code on two planned clips and two acceptance clips; addendum 2 lists the read-out settings that were chosen with that pilot in view. Before addendum 3: a CPU layout test that was not scored, and several passes of the tiny test chain on CPU with

Whisper small.en; in the passes before the code review, each made two runs on the B2 test item b2t00, and after it the tiny scale used a development item. Nobody listened to those clips; their transcripts were read only to check the code, and they were deleted. E4's results chose B2's guide range, k range and comparators. Before addendum 4: passes of the tiny test chain on CPU with Whisper small.en, each with 2 A0 runs and 2 C1 runs on the development item item01 (on the final code, two from a clean state and a third that was killed and resumed); nobody listened to those clips, only the logged measures were read, and the audio was deleted. Run C's recipes, arms and power simulation come from B2's results, and its 96 lines were written after Run B.

SA.3 Deviations and errata

- E5 v1: the K1 rule named flute, synth voice and clarinet captions, but the pilot compared three flute captions; this was not disclosed before the run. Addendum 1 says the rename came before any confirmatory run, which is wrong: the v1 grid had finished at 19:00:24Z, and v1 used the code with the registered hashes. The v1 scorer did not compute several registered quantities.

- E5 v2: the first K0 run failed and the melodies were rebuilt under the registered K0 clause, with the remedy chosen by the author after the K0 result and before any v2 model output (Section S5). The first start of the K0 rerun was stopped by a tool time limit before it wrote any K0 file; its partial renders were deleted and it was started again unchanged, as addendum 2 records. The run state was not on durable storage; there was one pass and one attempt. Null gaps were not silent. Some result files have no `git_commit` field; they are tied to the run by hashes, and every file that has the field names `83d46e7`. The per-job check did not re-test the decode settings; the ledgers record decode chunk 256 with no CPU or out-of-memory decode in all 2496 jobs.

- B1: the result file's label "timing not moved in 9 items" is a fixed phrase; 10 of the 12 lines were not improved.

- B2: the A0 time projection was about six times too long, and the registered trim removed clarinet at $k = 3$ and the no-guide arm. `b2_selection.json` marks every unclocked recipe as eligible, including two below the 0.60 gate rate; the rule puts no gate on B^*, so this is a naming issue. ACE-Step copied a configuration file into the checkpoint

that is not in the registered hash list; the model file it accompanies matches its registered hash. Ledger rows carry no time stamps; the log lines show the selection was written before Stage T began. Audio and latent files were kept outside the result package, so their hashes could not be checked again after the run. The registered word matcher is not always the longest in-order match (Section S2).

- Run C: three Colab sessions touched the run state. The first ran setup, A0, all 1536 generations and scores, decide and package; a second on the same runtime and a third on a new one ran only setup, decide and package, generated and scored nothing, reused A0 on the same runtime fingerprint and printed the same verdict. The reason for the second session is not recorded. The registered files come from the third. The author re-zipped the result package before sending it and moved 140 of the 192 labelled demo clips into a separate folder to send them apart, without listening to them. No run file records a hash of the ledgers or the result files, so their integrity after the re-zip rests on cross-checks: every kept draw and every J recomputed from the ledger rows matches. The package pins are recorded only as empty lists of mismatches, not as the installed versions, and the demo MP3s cannot be tied byte for byte to their WAVs, which were kept outside the package.

- Whisper output: some transcripts in the B1, B2 and C1 result files are known hallucinated credit lines. They are kept unedited so the files match the run; in B2, 35 of the 39 such clips count as wordless, and in C1, as in B2, the registered scorer counts a transcript made only of the registered hallucination strings and vowel sounds as wordless.

SB Full tables

E5 v2 quality pass rates by condition: lit 0.875, ctl 0.778, rel+2 0.771, rel+5 0.778, rel−3 0.771, seq 0.8125, ana 0.799 (144 draws each). H4 classification counts in seq (of 144 draws): T_6A 0, T_5A 0, T_7A 0, T_4A 5, T_2A 19, D 6, none 114, tie 0.

Paired contrasts of S^* against each arm, with 95% intervals from the registered item bootstrap: mean kept q +0.131 [0.064, 0.200] against S^*_{b0}, +0.164 [0.092, 0.240] against S^*_{rand}, +0.126 [0.049, 0.205] against S^*_{shuf} and +0.342 [0.246, 0.433] against the guide at $k = 4$; ΔJ +0.25 [0, 0.5] against S^*_{b0} and S^*_{rand}, +0.29 [0.08, 0.50] against S^*_{shuf} and +0.46 [0.21, 0.71] against the guide at $k = 4$.

Table 6: E5 v2, every registered index and difference, mean [95% cluster-bootstrap interval over 36 melodies]. ALL: 144 draws per condition. PASS: gate-passing draws (lit 126, ctl 112, rel+2 111, rel+5 112, rel−3 111, ana 115). _ctl: the same index computed in the ctl condition. Source: `results/gpu_run2/e5v2_main.json`.

Index	ALL	PASS
CI(lit)	0.471 [0.405, 0.536]	0.498 [0.426, 0.570]
CI(ctl)	0.106 [0.063, 0.152]	0.095 [0.043, 0.150]
CI(null)	−0.025 [−0.044, −0.007]	–
RI_{+2}	−0.021 [−0.043, 0.001]	−0.012 [−0.041, 0.021]
LI_{+2}	0.131 [0.076, 0.193]	0.136 [0.077, 0.198]
RI_{+2}_ctl	−0.010 [−0.033, 0.011]	−0.012 [−0.040, 0.018]
RI_{+5}	0.002 [−0.024, 0.028]	0.012 [−0.017, 0.041]
LI_{+5}	0.098 [0.051, 0.151]	0.098 [0.048, 0.152]
RI_{+5}_ctl	−0.015 [−0.029, −0.001]	−0.007 [−0.025, 0.012]
RI_{-3}	0.020 [−0.002, 0.041]	0.025 [−0.001, 0.049]
LI_{-3}	0.103 [0.057, 0.150]	0.106 [0.048, 0.163]
RI_{-3}_ctl	−0.008 [−0.024, 0.010]	−0.004 [−0.024, 0.018]
ANA	0.008 [−0.018, 0.034]	0.007 [−0.025, 0.038]
CI(lit) − CI(ctl) (H2)	0.365 [0.306, 0.425]	0.403 [0.327, 0.479]
RI_{+2} − RI_{+2}_ctl	−0.011 [−0.034, 0.014]	0.000 [−0.028, 0.028]
RI_{+5} − RI_{+5}_ctl	0.017 [−0.009, 0.042]	0.019 [−0.013, 0.051]
RI_{-3} − RI_{-3}_ctl	0.028 [0.005, 0.050]	0.029 [0.003, 0.054]
RI_{+2} − LI_{+2}	−0.152 [−0.221, −0.089]	−0.148 [−0.221, −0.074]
RI_{+5} − LI_{+5}	−0.096 [−0.158, −0.040]	−0.086 [−0.153, −0.023]
RI_{-3} − LI_{-3}	−0.082 [−0.136, −0.030]	−0.081 [−0.146, −0.017]

Table 7: B1, per line: timing error without the hook (median seconds between Whisper's word onsets and the written onsets) and the reduction in each clocked arm. A reduction of at least 0.30 counts as improved. 3 draws per line and arm. Source: `results/gpu_run3/b1_result.json`.

Line	No hook (s)	A, β 1	A, β 2	A, β 4	Random, β 2
item00	4.88	0.05	0.07	0.89	−0.02
item01	3.21	0.05	0.13	−0.02	0.01
item02	5.90	0.00	0.41	0.06	0.06
item03	2.72	0.71	0.77	0.84	0.01
item04	3.95	−0.06	−0.01	0.37	0.00
item05	4.11	0.10	0.03	0.36	0.01
item06	3.69	−0.08	−0.08	0.04	0.00
item07	2.96	0.00	−0.10	−0.13	0.00
item08	2.46	−0.60	−0.02	−1.11	−0.12
item09	5.00	0.00	0.13	0.12	0.00
item10	2.47	0.11	0.28	0.36	−0.01
item11	2.34	0.01	−0.01	0.33	0.00
Lines improved		1	2	6	0
Median reduction		0.003	0.047	0.229	−0.001
Word recall (36 clips)	0.595	0.585	0.600	0.525	0.586
Vocal-gate rate	0.944	0.972	0.972	0.972	0.944

SC Compute

CPU work (measurement tests, the E5 ruler and K0 checks, unit tests and the small end-to-end test chains) ran on a 4-core x86 machine with 15 GB of RAM and no GPU. E0 took 668 s (`results/cpu/e0_ruler.json`). All GPU work ran on one NVIDIA L4 (24 GB) in a hosted notebook. Installation and model download times were not recorded.

In GPU run 1, scoring took longer than generating: E4 scoring with Whisper, SwiftF0 and Basic Pitch took 4092 s (`results/gpu_run1/e4_sweep.json`), while a warm generation of a 14 s clip took about 2 to 3 s. In Run A, the setup stage ran from 09:43:35 to 09:58:57 (`e5v2_setup.json`) and the E6 read-out finished at 12:36:06 (`e6_gate.json`). The Run B stage minutes are in `results/gpu_run3/status.json`; the run log gives measured costs of 8.7 s per text-to-music run and 10.3 and 12.1 s per guided run in Stages D and T, generation and scoring together. The Run C stage minutes are in `results/gpu_run4/status.json`, summed over its three sessions; that file was written while the last session was still packaging, so the last packaging time is missing. In total the GPU work took about 16 hours of L4 time, not counting installation.

Table 8: B2 Stage T, every arm (24 melodies × 4 draws = 96 clips per arm). S^*_{b0} is also B^* and the E4 setting at $k = 3$. Rates and means are over the 96 clips unless the row says otherwise. Source: `results/gpu_run3/b2_result.json`; the last three rows are post-hoc checks computed from its clip rows by `experiments/checks/runb_secondary.py` and stored in `results/checks/runb_secondary.json` (WR chance scores each transcript against the other 23 lyrics with `dacapo/words.py`).

Measure	S^*	S^*_{b0}	S^*_{rand}	S^*_{shuf}	$k = 4$
J (melodies of 24)	14	8	8	7	3
Strict melodies	2	2	1	1	0
Joint clips (of 96)	31	13	11	9	5
Strict clips (of 96)	2	3	1	1	0
Mean kept q (best of 4)	0.517	0.385	0.353	0.391	0.175
Mean q per clip	0.341	0.201	0.169	0.226	0.064
J, best of 1	0.32	0.14	0.11	0.09	0.05
J, best of 2	0.47	0.23	0.21	0.17	0.09
J, best of 3	0.53	0.29	0.28	0.24	0.11
NPA_v	0.439	0.446	0.438	0.408	0.750
NPA_v, chance	0.103	0.096	0.095	0.096	0.108
NPA_v, octave-tolerant	0.558	0.500	0.500	0.480	0.786
NPA on the mix	0.437	0.483	0.488	0.439	0.841
WR (Whisper, in order)	0.495	0.322	0.281	0.310	0.072
WR, any order	0.500	0.324	0.287	0.380	0.072
WER, capped at 1	0.566	0.704	0.738	0.838	0.933
Recall, first half	0.50	0.32	0.29	0.30	0.10
Recall, second half	0.49	0.33	0.28	0.32	0.05
Recall, last word	0.26	0.22	0.17	0.33	0.02
Vocal-gate rate	0.990	0.948	0.969	1.000	0.896
Vowel-only rate	0.000	0.115	0.104	0.000	0.115
Wordless rate	0.021	0.281	0.312	0.031	0.708
Instrument bleed rate	0.042	0.073	0.073	0.052	0.156
Guide-copy index	0.415	0.420	0.411	0.396	0.601
Onset error, median (s)	0.42	1.73	1.86	4.49	3.75
Onset error, signed (s)	−0.38	−1.73	−1.86	−4.44	−3.75
Clips with onset error	71	46	39	50	12
CTC WR	0.179	0.160	0.121	0.128	0.045
J with CTC WR (kept draws)	2	3	1	2	1
Post-hoc and exploratory, from the clip rows (`results/checks`):					
WR chance (other 23 lyrics)	0.145	0.092	0.086	0.113	0.026
J, words in any order	15	8	8	10	3
J with CTC WR, draws re-selected by CTC	4	3	3	2	1

Table 9: B2 Stage D, the full development table (12 E4 lines × 3 draws = 36 clips per configuration), in the order of the registered selection key. J_{dev}: lines whose best of 3 draws is joint. Best of 3: mean over lines of the best q. Gate: vocal-gate rate. Eligible: a clocked configuration with a vocal-gate rate of at least 0.60. Configurations are named guide, k and clock (A: alignment heads, W: all 80 heads of layers 2–6, followed by β). Source: `results/gpu_run3/b2_selection.json`.

Configuration	J_{dev}	Best of 3	Mean q	Gate	Eligible
`oohs_k3_A8` (S^*)	9	0.553	0.407	0.972	yes
`oohs_k4_A8`	6	0.558	0.356	1.000	yes
`oohs_k3_A4`	6	0.435	0.289	0.944	yes
`oohs_k3_A2`	5	0.411	0.237	0.944	yes
`oohs_k3_W2`	5	0.387	0.218	0.889	yes
`oohs_k4_A4`	4	0.280	0.189	0.944	yes
`clarinet_k4_A8`	3	0.251	0.118	0.694	yes
`clarinet_k4_A4`	3	0.168	0.070	0.528	no
`oohs_k3_none` (B^*)	2	0.241	0.116	0.889	–
`oohs_k5_A8`	2	0.208	0.139	0.944	yes
`oohs_k4_none`	2	0.186	0.078	0.917	–
`oohs_k4_A2`	2	0.185	0.094	0.889	yes
`oohs_k4_W2`	2	0.172	0.092	0.889	yes
`clarinet_k4_W2`	2	0.121	0.049	0.361	no
`clarinet_k4_A2`	2	0.115	0.045	0.500	no
`clarinet_k4_none`	1	0.080	0.035	0.333	–
`clarinet_k5_A8`	1	0.071	0.024	0.583	no
`oohs_k5_A4`	1	0.069	0.023	0.972	yes
`clarinet_k5_none`	0	0.059	0.024	0.528	–
`oohs_k5_W2`	0	0.052	0.022	0.972	yes
`clarinet_k5_W2`	0	0.046	0.015	0.472	no
`oohs_k5_A2`	0	0.046	0.020	0.972	yes
`clarinet_k5_A2`	0	0.020	0.007	0.528	no
`oohs_k5_none`	0	0.020	0.011	0.972	–
`clarinet_k5_A4`	0	0.000	0.000	0.528	no

Table 10: GPU runs. Minutes are from the run log and status files; the run log is the only source for GPU run 1's total.

Run	Stages	Minutes
GPU run 1 (2026-10-26)	E1, E1b, E3, E4, E5 v1, smoke and a cappella tests (7 of 8 stages)	about 144
Run A (2026-10-28)	setup 15.4, A0 1.6, Phase 0b 71.1, E5 grid with quality and scoring 65.1, E6 read-out 19.3, packaging 0.8	about 173
Run B (2026-10-29/30)	setup 0.9, A0 4.5, B1 26.0, Stage D 154.0, selection 0.2, Stage T 96.9, decide 0.6	283.1
Run C (2026-10-30/31)	setup 3.7, A0 1.8, C1 generation and scoring 322.4, decide 6.6, packaging 4.8 (three sessions)	$\geq$ 339.3

SD Reproducibility

Code, result files, registrations and figures are at https://github.com/ninondev/DaCapo under the MIT license. Every number in this paper comes from a file in `results/`, from a registration file, or, where marked, from the run log `LOG.md`; `figures/make_figures.py` draws every figure from the result files. The checks of Run B made after the run are in `results/checks/runb_secondary.json`, computed only from `results/gpu_run3/b2_result.json` by `experiments/checks/runb_secondary.py`; those of Run C are in `results/checks/runc_secondary.json`, computed only from `results/gpu_run4/c1_result.json` and `results/gpu_run3/b2_result.json` by `experiments/checks/runc_secondary.py`. The notebooks `notebook.ipynb` (Run A), `notebook_runB.ipynb` (Run B) and `notebook_runC.ipynb` (Run C) install the pinned packages, download the weights and run every stage; `experiments/run_all.py` lists the command of each earlier step.

Commits. `1c91106` (addendum 2, step 1: K0 remedy), `b6ed62f` (K0 rerun and addendum 2 complete), `f40132d` (E7 Stage S registered), `83d46e7` (Run A code), `6835deb` (Run A results), `680b24b` (B1 code addendum), `bd61a9f` (addendum 3 and Run B code), `00d47db` (Run B results), `0c97266` (addendum 4 and Run C code), `79e2868` (listening plan of the Run C pack), `0dfeb0e` (Run C results), `97f3a19` (E1b figure title), `44039fc` and `9ddcdbc` (README wording), `7527e13` (paper figures without titles). This version of the paper describes the repository at commit `7527e13`.

Pinned versions. ACE-Step 1.5 at commit `ca1e85fe9430179831e6bc6be790c332190a3866`, weights at revision `19671f406d603126926c1b7e2adc169acbcade22` (turbo model file SHA-256 `3f6e0797fad420a39bd33979eb6e840e30989e34a3794e843d23b60ec6e422d7`). Whisper large-v3 at revision `06f233fe06e710322aca913c1bc4249a0d71fce1`. The CTC recogniser is wav2vec2-large-960h-lv60-self at revision `54074b1c16f4de6a5ad59affb4caa8f2ea03a119`. The forced aligner of the E6 read-out is torchaudio's `MMS_FA` model [Pratap et al., 2024], loaded without the star token; the notebook did not install torchaudio, so it used the copy preinstalled in the hosted runtime next to PyTorch 2.11.0, whose version was not recorded. FluidSynth 2.3.4 [FluidSynth developers, 2023] with FluidR3_GM [Wen, 2008] from the Debian and Ubuntu package `fluid-soundfont-gm` 3.1-5.3 (SHA-256 `74594e8f4250680adf590507a306655a299935343583256f3b722c48a1bc1cb0`). Measurement packages: swift-f0 0.3.0, basic-pitch 0.4.0 (ONNX), onnxruntime 1.31.0, librosa 1.0.0, mir_eval 0.8.2, pretty_midi 0.2.11.post0, demucs 4.1.0. GPU runtime: Python 3.13.16, PyTorch 2.11.0 with CUDA 13.0, transformers 4.57.6, diffusers 0.39.0, numpy 2.1.3.

Determinism. GPU runs use deterministic algorithms, cuDNN benchmark off, TF32 matrix multiplication off, a fixed cuBLAS workspace, and a VAE decode chunk of 256; the global random state is pinned before each run. Under these settings a repeated run and a run in a fresh process gave identical latents and audio on the L4. The same seed gives different audio on CPU and GPU.

Not in the repository. Audio, latents, run ledgers, the blind listening packs and their keys are kept outside the repository; the keys are archived privately and opened only after ratings close.

Who did what

The author led the project. The author set the research questions; searched for a model to study and chose ACE-Step 1.5 with help from a general-purpose AI assistant; decided to build melody control inside the renderer; required every main study to be preregistered and the final claims about how the outputs sound to rest on blind listening; ran every GPU experiment; and made the decisions that the registrations record as the author's, such as the remedy after the failed K0 check (Section S5), the GPU budget of Run B and the inclusion of head set W in its development grid. The use of generative AI is declared in the article.

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